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(54) **SYSTEM AND METHOD FOR DMRS
ANTENNA PORT INDICATION FOR URLLC**

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2020, now Pat. No. 12,035,327.

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H04L 5/00 (2006.01)

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CPC **H04W 72/23** (2023.01); **H04L 5/0051**
(2013.01); **H04L 5/0094** (2013.01)

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CPC H04W 72/23; H04L 5/0051; H04L 5/0094
See application file for complete search history.

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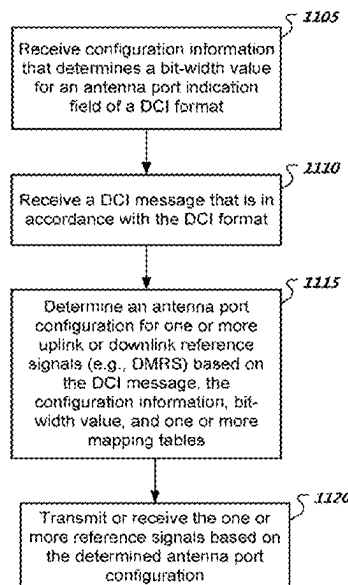
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(57) **ABSTRACT**

Systems, devices, and techniques for antenna port configura-
tion in wireless communication systems are described. A
described technique performed by a user equipment (UE)
includes receiving configuration information that determines
a bit-width value for an antenna port indication field of a
downlink control information (DCI) format; receiving a DCI
message that is in accordance with the DCI format; and
determining an antenna port configuration for one or more
demodulation reference signals based on the DCI message,
the configuration information, and the bit-width value.

20 Claims, 11 Drawing Sheets



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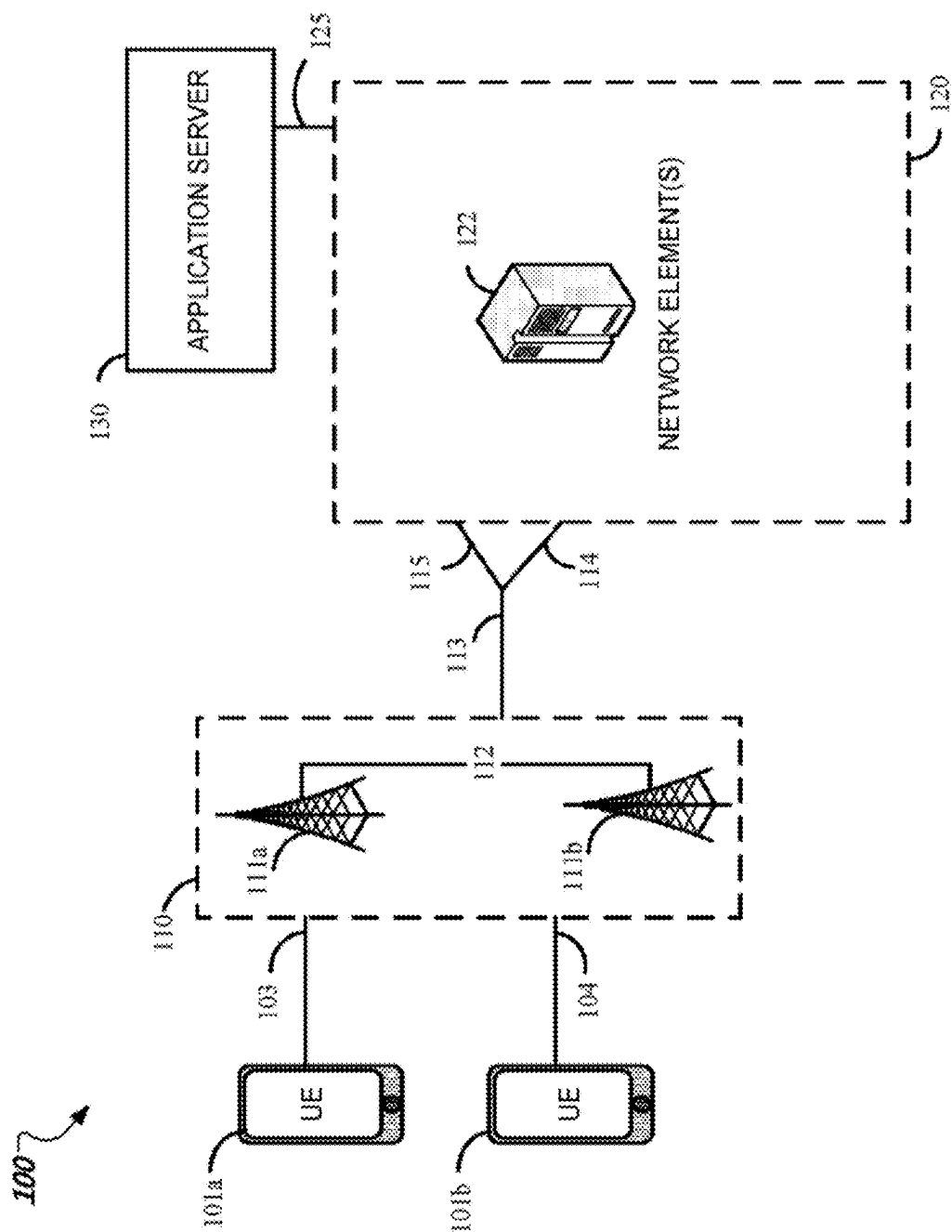


FIG. 1

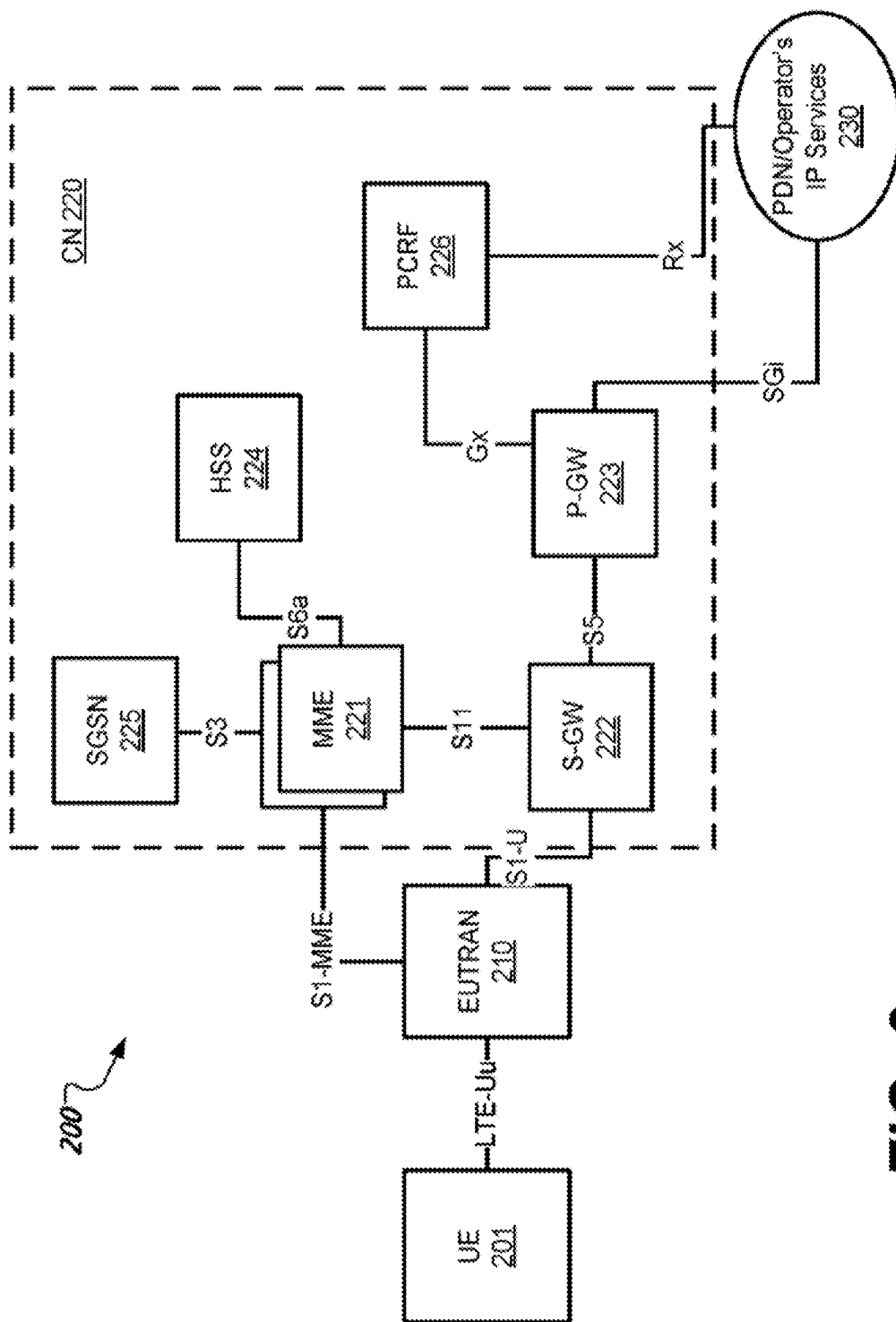
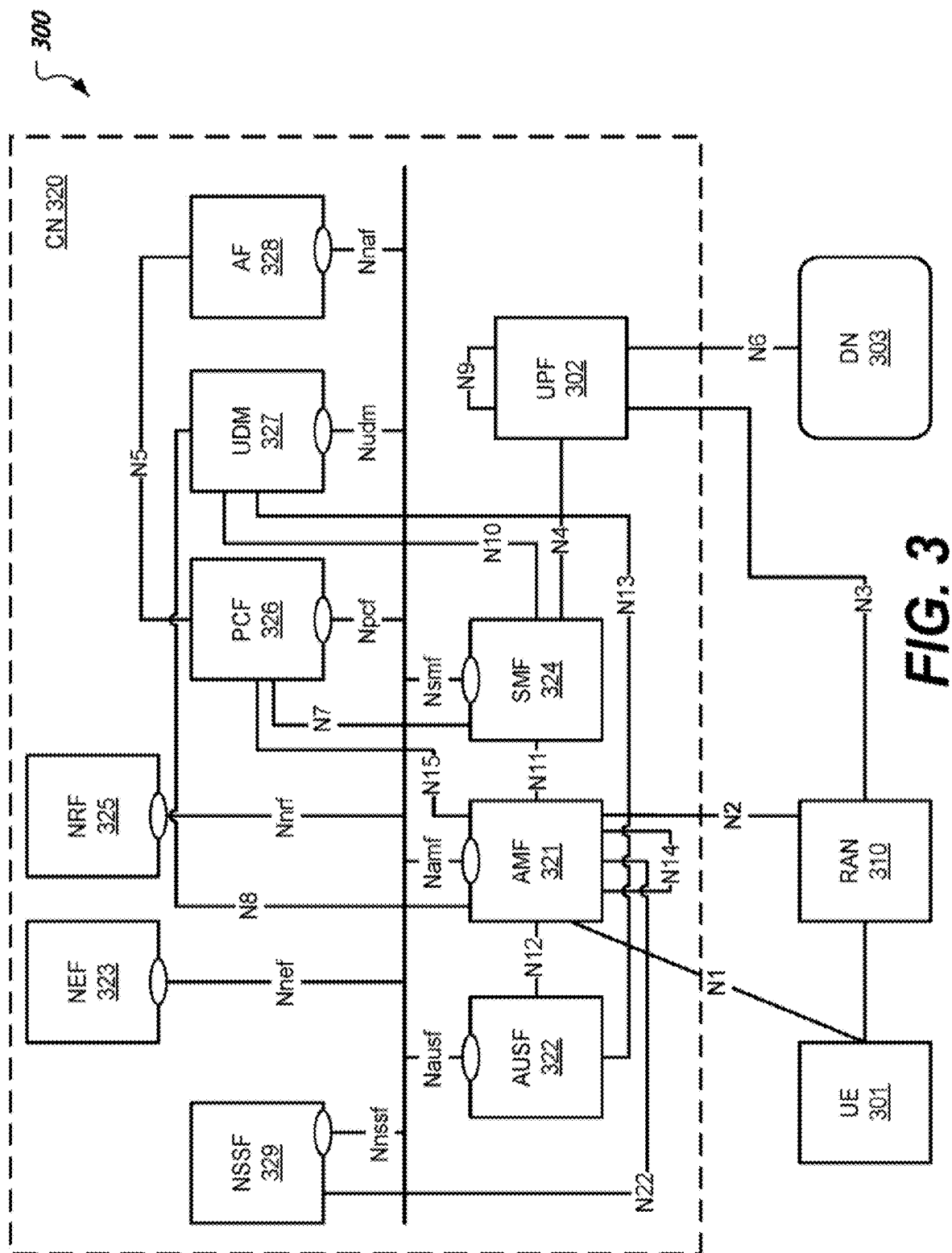
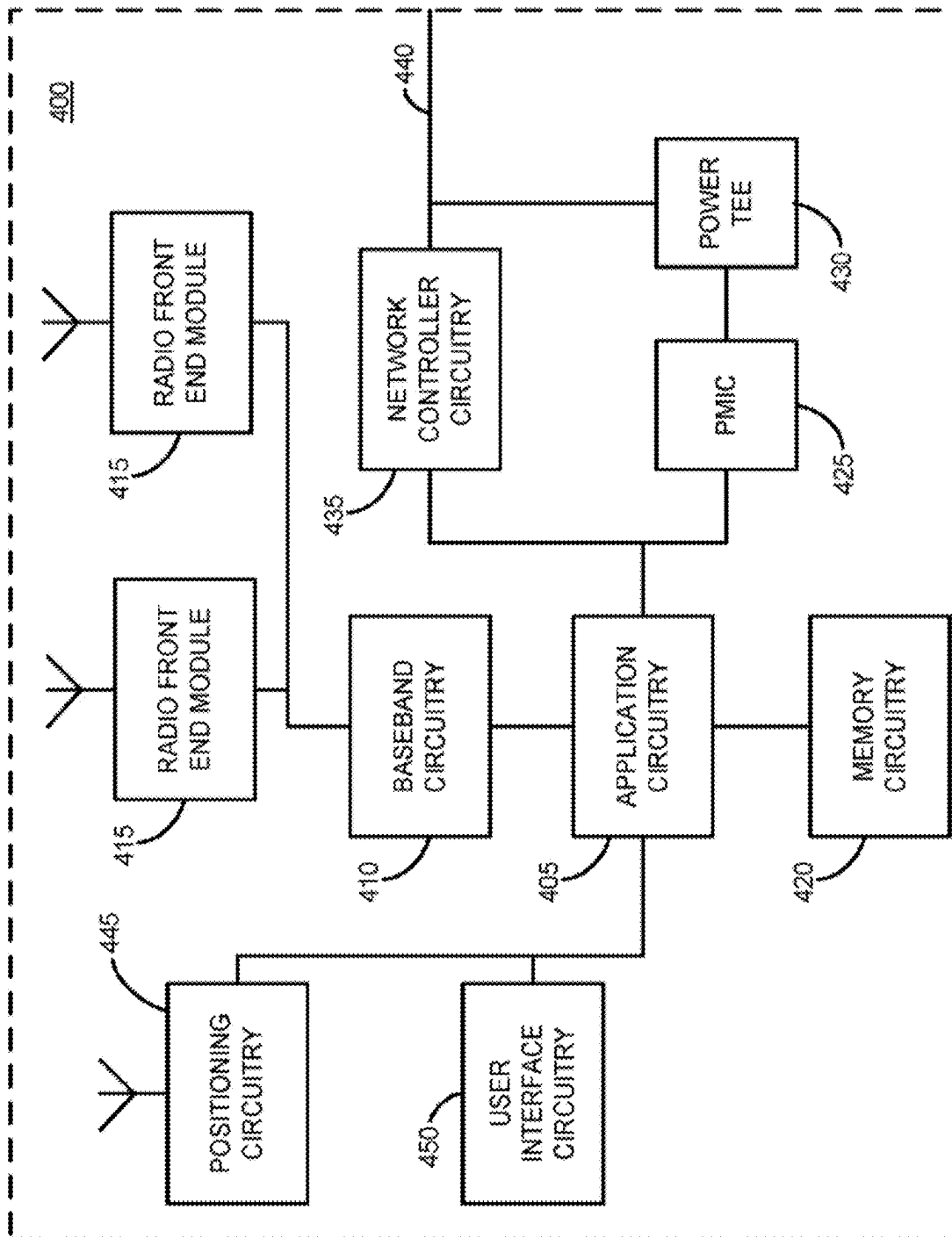


FIG. 2



**FIG. 4**

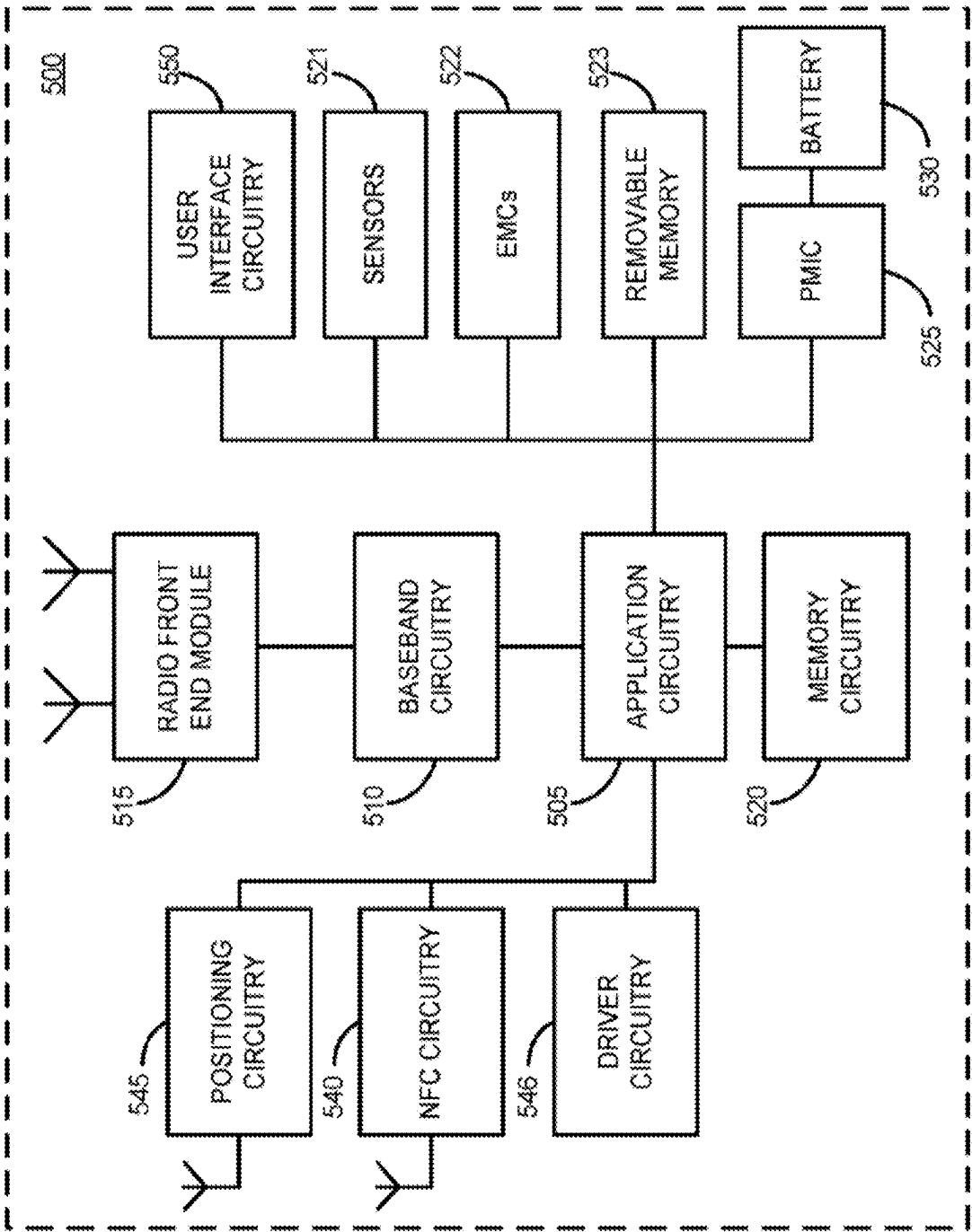
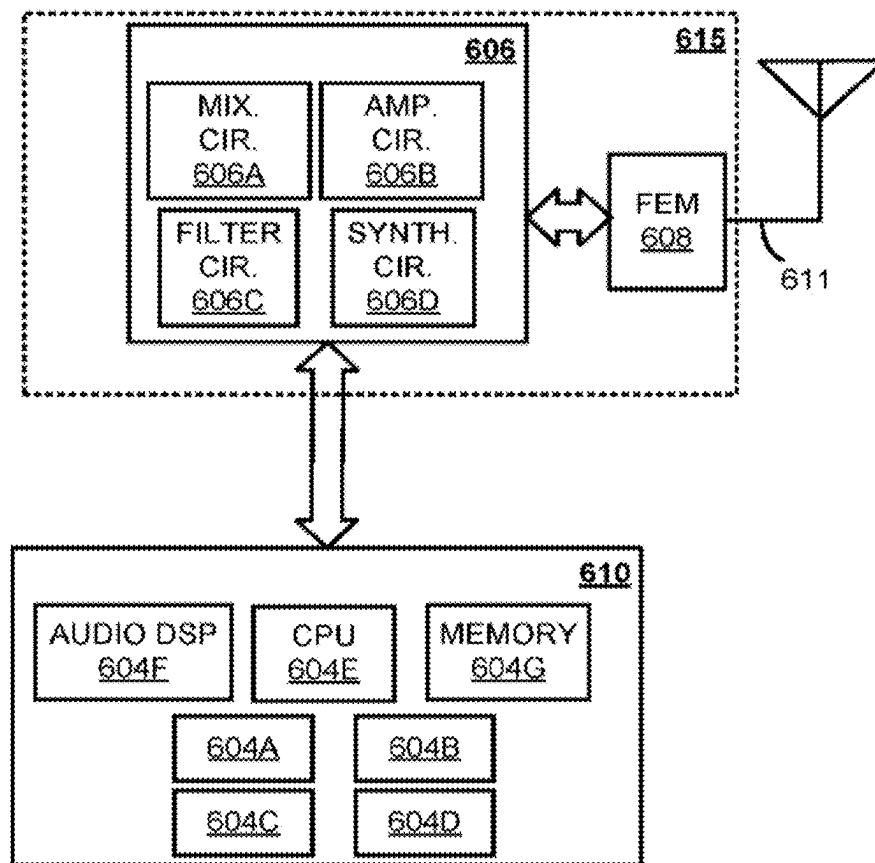
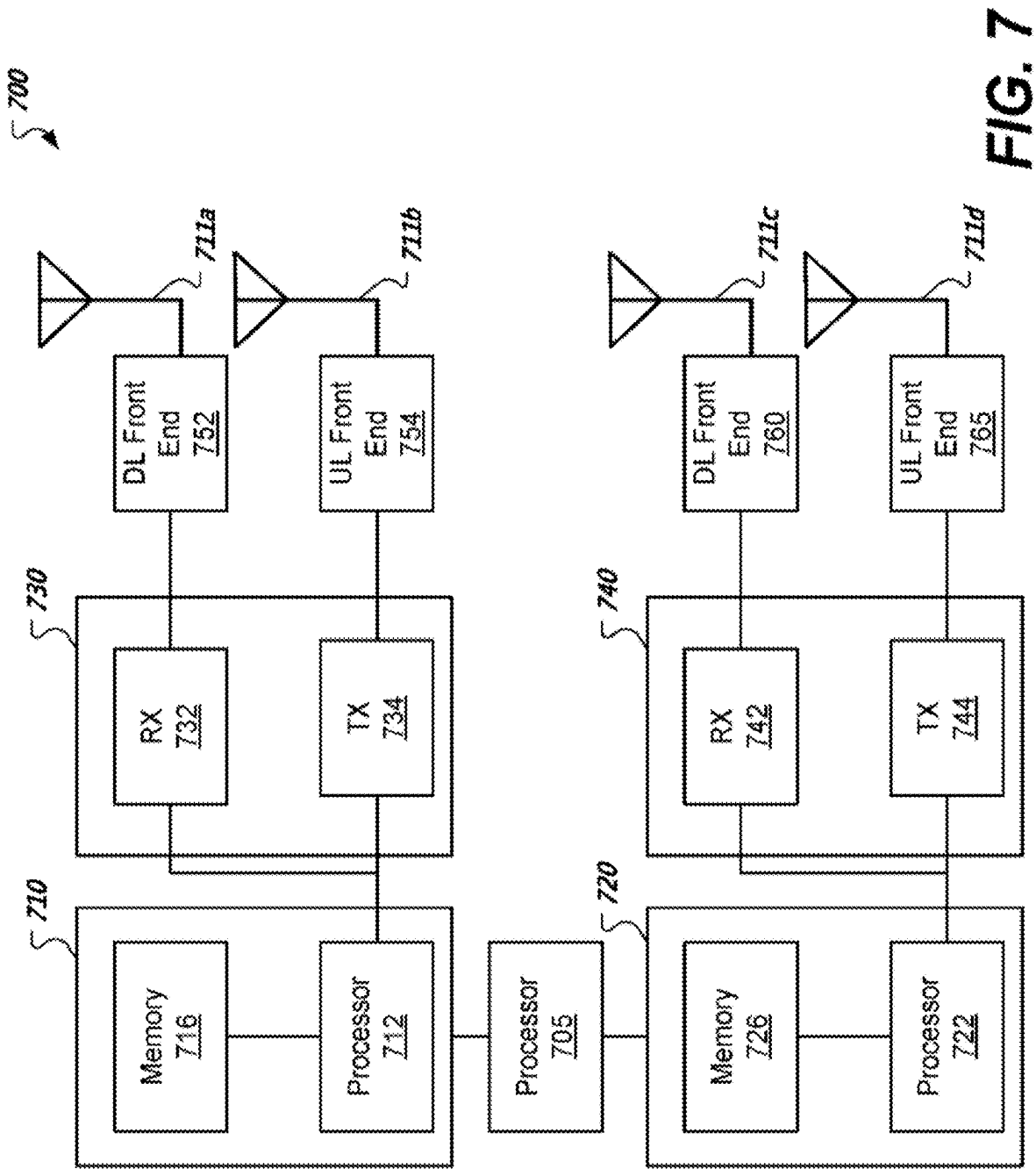
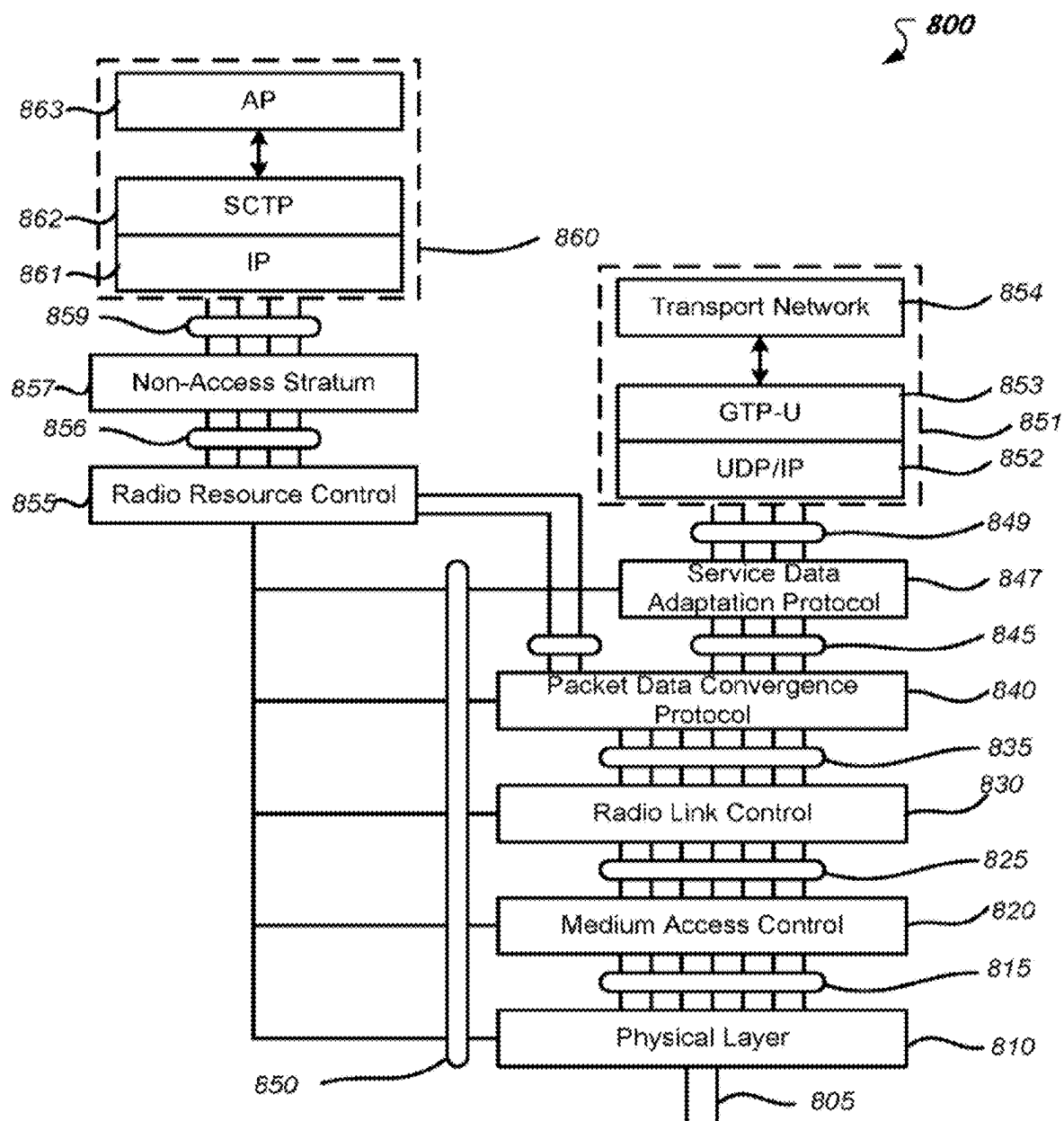


FIG. 5

**FIG. 6**



**FIG. 8**

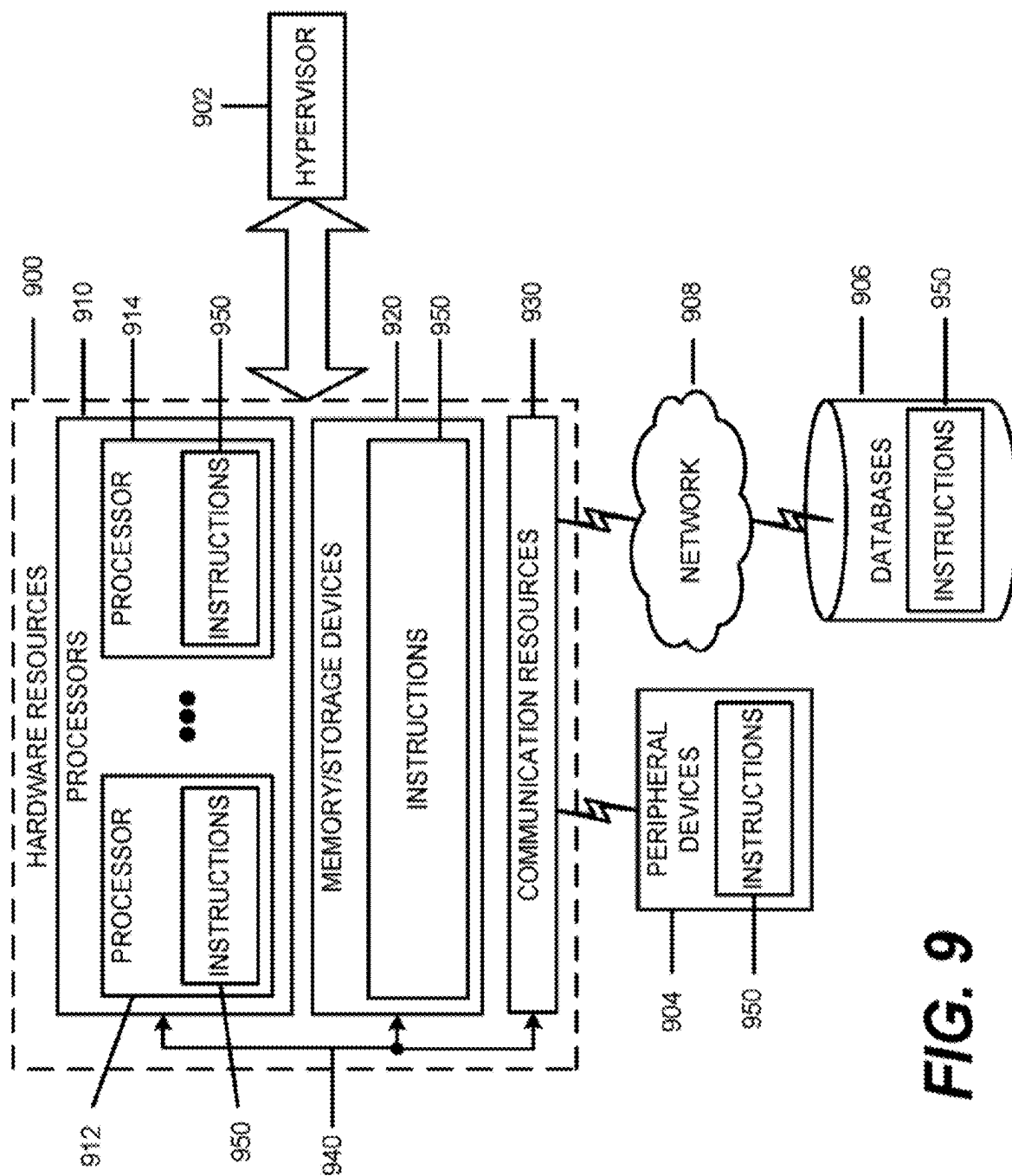


FIG. 9

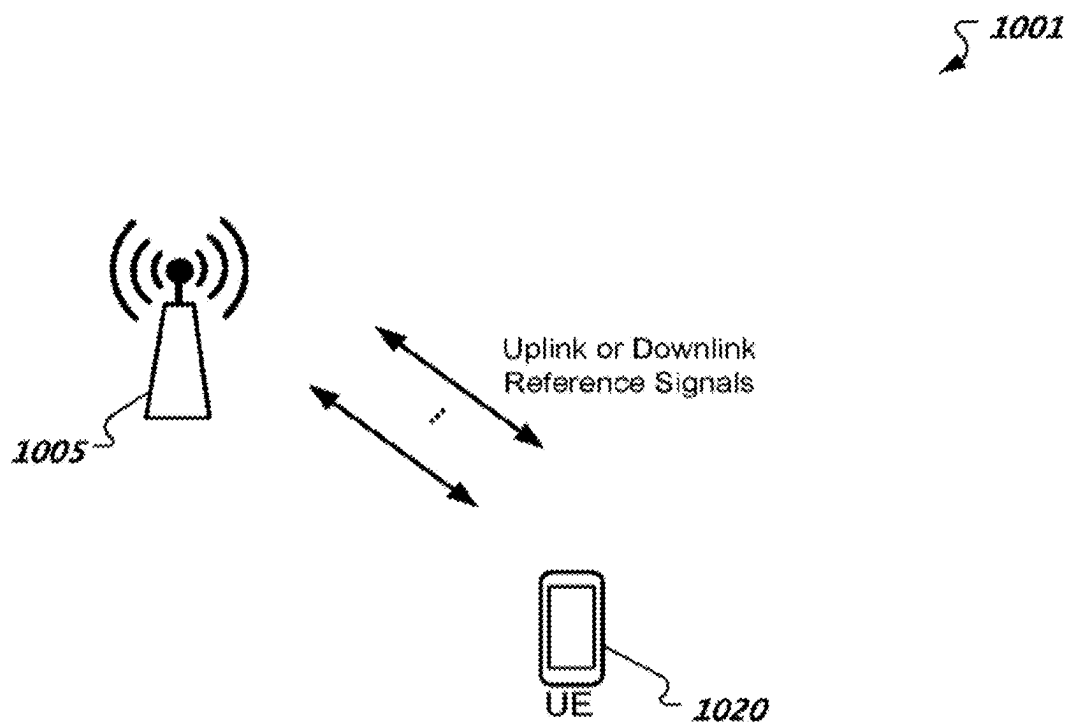
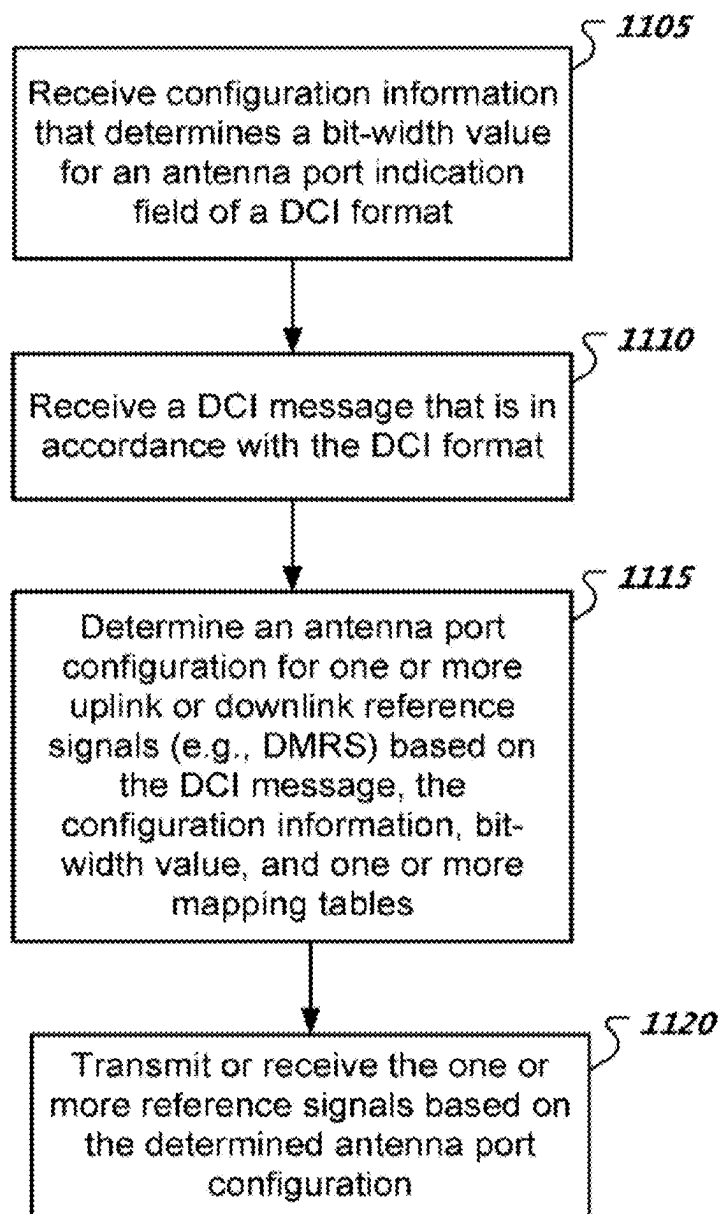


FIG. 10

**FIG. 11**

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SYSTEM AND METHOD FOR DMRS ANTENNA PORT INDICATION FOR URLLC

CROSS-REFERENCE TO RELATED APPLICATION

The present application is a Continuation of U.S. application Ser. No. 17/441,831, which is a U.S. National Phase application under 35 U.S.C. § 371 of International Patent Application No. PCT/US2020/031091, filed on May 1, 2020, which claims the benefit of the priority of U.S. Provisional Patent Application No. 62/843,273, entitled “SYSTEM AND METHOD FOR DMRS ANTENNA PORT INDICATION FOR URLLC” and filed on May 3, 2019. The above-identified applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

This disclosure relates generally to wireless communication systems.

BACKGROUND

Base stations, such as a node of radio access network (RAN), can wirelessly communicate with wireless devices such as user equipment (UE). A downlink (DL) transmission refers to a communication from the base station to the wireless device. An uplink (UL) transmission refers to a communication from the wireless device to another device such as the base station. Base stations can transmit control signaling in order to control wireless devices that operate within their network.

SUMMARY

Systems, devices, and techniques for antenna port configuration in wireless communication systems are described. A described technique performed by a user equipment (UE) includes receiving configuration information that determines a bit-width value for an antenna port indication field of a downlink control information (DCI) format; receiving a DCI message that is in accordance with the DCI format; and determining an antenna port configuration for one or more demodulation reference signals based on the DCI message, the configuration information, and the bit-width value. Other implementations include corresponding systems, apparatus, and computer programs to perform the actions of methods defined by instructions encoded on computer readable storage.

These and other implementations can include one or more of the following features. In some messages, the DCI message includes an antenna port indication. Determining the antenna port configuration can include determining whether there is an antenna port indication in the DCI message based on the bit-width value. Determining the antenna port configuration can include determining the antenna port configuration based on the antenna port indication and a mapping between antenna port indication values and antenna port numbers. In some implementations, the configuration information specifies a subset of the mapping between antenna port indication values and antenna port numbers. Determining the antenna port configuration can include using a predetermined antenna port configuration based on a determination that the bit-width value is a zero value, where the zero value indicates that the DCI message does not include the antenna port indication field.

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Receiving the configuration information can include receiving the configuration information via radio resource control (RRC) signaling. Implementations can include receiving the one or more demodulation reference signals based on the determined antenna port configuration. Implementations can include transmitting the one or more demodulation reference signals based on the determined antenna port configuration.

A UE can include one or more processors, circuitry such as a transceiver configured to communicate with a base station, and a memory storing instructions that, when executed by the one or more processors, cause the one or more processors to perform operations. The operations can include receiving, via the circuitry, configuration information that determines a bit-width value for an antenna port indication field of a DCI format; receiving, via the circuitry, a DCI message that is in accordance with the DCI format; and determining an antenna port configuration for one or more demodulation reference signals based on the DCI message, the configuration information, and the bit-width value.

Communication products such as a communication processor(s) can include circuitry configured to receive configuration information that determines a bit-width value for an antenna port indication field of a DCI format, and receive a DCI message that is in accordance with the DCI format; and circuitry configured to determine an antenna port configuration for one or more demodulation reference signals based on the DCI message, the configuration information, and the bit-width value. The communication processor(s) can include circuitry configured to transmit or receive the one or more demodulation reference signals based on the antenna port configuration. In some implementations, the DCI message includes an antenna port indication, and wherein the antenna port configuration is determined based on the antenna port indication and a mapping between antenna port indication values and antenna port numbers. In some implementations, the antenna port configuration is determined based on a predetermined antenna port configuration if the bit-width value is a zero value, and wherein the zero value indicates that the DCI message does not include the antenna port indication field. In some implementations, the configuration information is carried by RRC signaling.

The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features and advantages will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 illustrates an example of a wireless communication system.

FIG. 2 illustrates an example architecture of a system including a core network.

FIG. 3 illustrates another example architecture of a system including a core network.

FIG. 4 illustrates an example of infrastructure equipment.

FIG. 5 illustrates an example of a platform or device.

FIG. 6 illustrates example components of baseband circuitry and radio front end circuitry.

FIG. 7 illustrates example components of cellular communication circuitry.

FIG. 8 illustrates example protocol functions that may be implemented in wireless communication systems.

FIG. 9 illustrates an example of a computer system.

FIG. 10 illustrates a diagram of an example of a wireless communication system.

FIG. 11 illustrates an example of a process for determining an antenna port configuration.

Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

FIG. 1 illustrates an example of a wireless communication system 100. For purposes of convenience and without limitation, the example system 100 is described in the context of the LTE and NR communication standards as defined by the Third Generation Partnership Project (3GPP) technical specifications. However, other types of communication standards are possible. The system 100 can be configured for Ultra Reliable Low Latency Communications (URLLC), enhanced Mobile Broadband (eMBB), massive Machine Type Communications (mMTC), and/or others.

The system 100 includes UE 101a and UE 101b (collectively referred to as the “UEs 101”). In this example, the UEs 101 are illustrated as smartphones (e.g., handheld touchscreen mobile computing devices connectable to one or more cellular networks). In other examples, any of the UEs 101 may include other mobile or non-mobile computing devices, such as consumer electronics devices, cellular phones, smartphones, feature phones, tablet computers, wearable computer devices, personal digital assistants (PDAs), pagers, wireless handsets, desktop computers, laptop computers, in-vehicle infotainment (IVI), in-car entertainment (ICE) devices, an Instrument Cluster (IC), head-up display (HUD) devices, onboard diagnostic (OBD) devices, dashtop mobile equipment (DME), mobile data terminals (MDTs), Electronic Engine Management System (EEMS), electronic/engine control units (ECUs), electronic/engine control modules (ECMs), embedded systems, microcontrollers, control modules, engine management systems (EMS), networked or “smart” appliances, machine-type communications (MTC) devices, machine-to-machine (M2M) devices, Internet of Things (IoT) devices, or combinations of them, among others.

In some implementations, any of the UEs 101 may be IoT UEs, which can include a network access layer designed for low-power IoT applications utilizing short-lived UE connections. An IoT UE can utilize technologies such as M2M or MTC for exchanging data with an MTC server or device using, for example, a public land mobile network (PLMN), proximity services (ProSe), device-to-device (D2D) communication, sensor networks, IoT networks, or combinations of them, among others. The M2M or MTC exchange of data may be a machine-initiated exchange of data. An IoT network describes interconnecting IoT UEs, which may include uniquely identifiable embedded computing devices (within the Internet infrastructure), with short-lived connections. The IoT UEs may execute background applications (e.g., keep-alive messages or status updates) to facilitate the connections of the IoT network.

The UEs 101 are configured to connect (e.g., communicatively couple) with RAN 110. The RAN 110 can include one or more RAN nodes 111a and 111b (collectively referred to as “RAN nodes 111” or “RAN node 111”). In some implementations, the RAN 110 may be a next generation RAN (NG RAN), an evolved UMTS terrestrial radio access network (E-UTRAN), or a legacy RAN, such as a UMTS terrestrial radio access network (UTRAN) or a GSM EDGE radio access network (GERAN). As used herein, the term “NG RAN” may refer to a RAN 110 that operates in a 5G NR system 100, and the term “E-UTRAN” may refer to a RAN 110 that operates in an LTE or 4G system 100.

To connect to the RAN 110, the UEs 101 utilize connections (or channels) 103 and 104, respectively, each of which may include a physical communications interface or layer, as described below. In this example, the connections 103 and 104 are illustrated as an air interface to enable communicative coupling, and can be consistent with cellular communications protocols, such as a global system for mobile communications (GSM) protocol, a code-division multiple access (CDMA) network protocol, a push-to-talk (PTT) protocol, a PTT over cellular (POC) protocol, a universal mobile telecommunications system (UMTS) protocol, a 3GPP LTE protocol, a 5G NR protocol, or combinations of them, among other communication protocols.

The RAN 110 can include one or more RAN nodes 111a and 111b (collectively referred to as “RAN nodes 111” or “RAN node 111”) that enable the connections 103 and 104. As used herein, the terms “access node,” “access point,” or the like may describe equipment that provides the radio baseband functions for data or voice connectivity, or both, between a network and one or more users. These access nodes can be referred to as base stations (BS), gNodeBs, eNodeBs, eNBs, NodeBs, RAN nodes, road side units (RSUs), and the like, and can include ground stations (e.g., terrestrial access points) or satellite stations providing coverage within a geographic area (e.g., a cell), among others. As used herein, the term “NG RAN node” may refer to a RAN node 111 that operates in a 5G NR system 100 (for example, a gNB), and the term “E-UTRAN node” may refer to a RAN node 111 that operates in an LTE or 4G system 100 (e.g., an eNB). In some implementations, the RAN nodes 111 may be implemented as one or more of a dedicated physical device such as a macrocell base station, or a low power (LP) base station for providing femtocells, picocells or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells.

The RAN nodes 111 and the UEs 101 can be configured for multiple-input and multiple-output (MIMO) communications, including single or multi-beam communications. For example, a UE 101 can receive transmissions from one RAN node 111 at a time or from multiple RAN nodes 111 at the same time. The RAN nodes 111 and the UEs 101 can use beamforming for the UL, DL, or both. For example, one or more RAN nodes 111 can transmit (Tx) a beam towards a UE 101, and the UE 101 can receive data via one or more receive (Rx) beams at the same time. In some implementations, each of the RAN nodes 111 can be configured as a transmission and reception point (TRP). The RAN 110 can provide high-layer signaling for configuring beamforming such as by providing transmission configuration indication (TCI) state configuration information.

Any of the RAN nodes 111 can terminate the air interface protocol and can be the first point of contact for the UEs 101. In some implementations, any of the RAN nodes 111 can fulfill various logical functions for the RAN 110 including, but not limited to, radio network controller (RNC) functions such as radio bearer management, uplink and downlink dynamic radio resource management and data packet scheduling, and mobility management.

In some implementations, the UEs 101 can be configured to communicate using orthogonal frequency division multiplexing (OFDM) communication signals with each other or with any of the RAN nodes 111 over a multicarrier communication channel in accordance with various communication techniques, such as, but not limited to, OFDMA communication techniques (e.g., for downlink communications) or SC-FDMA communication techniques (e.g., for uplink communications), although the scope of the tech-

niques described here not limited in this respect. The OFDM signals can comprise a plurality of orthogonal subcarriers.

In some implementations, a downlink resource grid can be used for downlink transmissions from any of the RAN nodes **111** to the UEs **101**, while uplink transmissions can utilize similar techniques. The grid can be a frequency grid or a time-frequency grid, which is the physical resource in the downlink in each slot. Such a time-frequency plane representation is a common practice for OFDM systems, which makes it intuitive for radio resource allocation. Each column and each row of the resource grid corresponds to one OFDM symbol and one OFDM subcarrier, respectively. The duration of the resource grid in the time domain corresponds to one slot in a radio frame. The smallest time-frequency unit in a resource grid is denoted as a resource element (RE). Each resource grid can include a number of resource blocks, which describe the mapping of certain physical channels to resource elements. A resource block (RB) can include a collection of resource elements; in the frequency domain, this may represent the smallest quantity of resources that currently can be allocated. Physical downlink and uplink channels can be conveyed using such resource blocks. In some cases, a RB can be referred to as a physical resource block (PRB).

The RAN nodes **111** can transmit to the UEs **101** over one or more DL channels. Various examples of DL communication channels include a physical broadcast channel (PBCH), physical downlink control channel (PDCCH), and physical downlink shared channel (PDSCH). The PDSCH can carry user data and higher-layer signaling to the UEs **101**. Other types of downlink channels are possible. The UEs **101** can transmit to the RAN nodes **111** over one or more UL channels. Various examples of UL communication channels include physical uplink shared channel (PUSCH), physical uplink control channel (PUCCH), and physical random access channel (PRACH). Other types of uplink channels are possible. Devices such as the RAN nodes **111** and the UEs **101** can transmit reference signals. Examples of reference signals include a sounding reference signal (SRS), channel state information reference signal (CSI-RS), demodulation reference signal (DMRS or DM-RS), and phase tracking reference signal (PTRS). Other types of reference signals are possible.

A channel such as PDCCH can convey scheduling information of different types for one or more downlink and uplink channels. Scheduling information can include downlink resource scheduling, uplink power control instructions, uplink resource grants, and indications for paging or system information. The RAN nodes **111** can transmit one or more downlink control information (DCI) messages on the PDCCH to provide scheduling information, such as allocations of one or more PRBs. In some implementations, a DCI message transports control information such as requests for aperiodic CQI reports, UL power control commands for a channel, and a notification for a group of UEs **101** of a slot format. Downlink scheduling (e.g., assigning control and shared channel resource blocks to the UE **101b** within a cell) may be performed at any of the RAN nodes **111** based on channel quality information fed back from any of the UEs **101**. The downlink resource assignment information may be sent on the PDCCH used for (e.g., assigned to) each of the UEs **101** or a group of UEs.

In some implementations, the PDCCH carries information about the transport format and resource allocations related to the PDSCH channel, among other things. It may also inform the UEs **101** about the transport format, resource allocation,

and hybrid automatic repeat request (HARQ) information for providing HARQ feedback on an uplink channel based on a PDSCH reception.

In some implementations, the PDCCH uses control channel elements (CCEs) to convey the control information. Before being mapped to resource elements, the PDCCH complex-valued symbols may first be organized into quadruplets, which may then be permuted using a sub-block interleaver for rate matching. In some implementations, each PDCCH may be transmitted using one or more of these CCEs, in which each CCE may correspond to nine sets of four physical resource elements collectively referred to as resource element groups (REGs). Four Quadrature Phase Shift Keying (QPSK) symbols may be mapped to each REG. The PDCCH can be transmitted using one or more CCEs, depending on the size of DCI and the channel condition. In some implementations, there can be four or more different PDCCH formats defined with different numbers of CCEs (e.g., aggregation level, $L=1, 2, 4, \text{ or } 8$).

Downlink and uplink transmissions can occur in one or more component carriers (CCs). One or more bandwidth part (BWP) configurations for each component carrier can be configured. In some implementations, a DL BWP includes at least one control resource set (CORESET). In some implementations, a CORESET includes one or more PRBs in a frequency domain, and one or more OFDM symbols in a time domain. In some implementations, channels such as PDCCH can be transmitted via one or more CORESETs, with each CORESET corresponding to a set of time-frequency resources. CORESET information can be provided to a UE **101**, and the UE **101** can monitor time-frequency resources associated with one or more CORESETs to receive a PDCCH transmission.

The RAN nodes **111** are configured to communicate with one another using an interface **112**. In examples, such as where the system **100** is an LTE system (e.g., when the core network **120** is an evolved packet core (EPC) network as shown in FIG. 2), the interface **112** may be an X2 interface **112**. The X2 interface may be defined between two or more RAN nodes **111** (e.g., two or more eNBs and the like) that connect to the EPC **120**, or between two eNBs connecting to EPC **120**, or both. In some implementations, the X2 interface may include an X2 user plane interface (X2-U) and an X2 control plane interface (X2-C). The X2-U may provide flow control mechanisms for user data packets transferred over the X2 interface, and may be used to communicate information about the delivery of user data between eNBs. For example, the X2-U may provide specific sequence number information for user data transferred from a master eNB to a secondary eNB; information about successful in sequence delivery of PDCP protocol data units (PDUs) to a UE **101** from a secondary eNB for user data; information of PDCP PDUs that were not delivered to a UE **101**; information about a current minimum desired buffer size at the secondary eNB for transmitting to the UE user data, among other information. The X2-C may provide intra-LTE access mobility functionality, including context transfers from source to target eNBs or user plane transport control; load management functionality; inter-cell interference coordination functionality, among other functionality.

In some implementations, such as where the system **100** is a 5G NR system (e.g., when the core network **120** is a 5G core network as shown in FIG. 3), the interface **112** may be an Xn interface **112**. The Xn interface may be defined between two or more RAN nodes **111** (e.g., two or more gNBs and the like) that connect to the 5G core network **120**, between a RAN node **111** (e.g., a gNB) connecting to the 5G

core network **120** and an eNB, or between two eNBs connecting to the 5G core network **120**, or combinations of them. In some implementations, the Xn interface may include an Xn user plane (Xn-U) interface and an Xn control plane (Xn-C) interface. The Xn-U may provide non-guaranteed delivery of user plane PDUs and support/provide data forwarding and flow control functionality. The Xn-C may provide management and error handling functionality, functionality to manage the Xn-C interface; mobility support for UE **101** in a connected mode (e.g., CM-CONNECTED) including functionality to manage the UE mobility for connected mode between one or more RAN nodes **111**, among other functionality. The mobility support may include context transfer from an old (source) serving RAN node **111** to new (target) serving RAN node **111**, and control of user plane tunnels between old (source) serving RAN node **111** to new (target) serving RAN node **111**. A protocol stack of the Xn-U may include a transport network layer built on Internet Protocol (IP) transport layer, and a GPRS tunneling protocol for user plane (GTP-U) layer on top of a user datagram protocol (UDP) or IP layer(s), or both, to carry user plane PDUs. The Xn-C protocol stack may include an application layer signaling protocol (referred to as Xn Application Protocol (Xn-AP or XnAP)) and a transport network layer that is built on a stream control transmission protocol (SCTP). The SCTP may be on top of an IP layer, and may provide the guaranteed delivery of application layer messages. In the transport IP layer, point-to-point transmission is used to deliver the signaling PDUs. In other implementations, the Xn-U protocol stack or the Xn-C protocol stack, or both, may be same or similar to the user plane and/or control plane protocol stack(s) shown and described herein.

The RAN **110** is shown to be communicatively coupled to a core network **120** (referred to as a “CN **120**”). The CN **120** includes one or more network elements **122**, which are configured to offer various data and telecommunications services to customers/subscribers (e.g., users of UEs **101**) who are connected to the CN **120** using the RAN **110**. The components of the CN **120** may be implemented in one physical node or separate physical nodes and may include components to read and execute instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium). In some implementations, network functions virtualization (NFV) may be used to virtualize some or all of the network node functions described here using executable instructions stored in one or more computer-readable storage mediums, as described in further detail below. A logical instantiation of the CN **120** may be referred to as a network slice, and a logical instantiation of a portion of the CN **120** may be referred to as a network sub-slice. NFV architectures and infrastructures may be used to virtualize one or more network functions, alternatively performed by proprietary hardware, onto physical resources comprising a combination of industry-standard server hardware, storage hardware, or switches. In other words, NFV systems can be used to execute virtual or reconfigurable implementations of one or more network components or functions, or both.

An application server **130** may be an element offering applications that use IP bearer resources with the core network (e.g., UMTS packet services (PS) domain, LTE PS data services, among others). The application server **130** can also be configured to support one or more communication services (e.g., VOIP sessions, PTT sessions, group communication sessions, social networking services, among others) for the UEs **101** using the CN **120**. The application server

130 can use an IP communications interface **125** to communicate with one or more network elements **122**.

In some implementations, the CN **120** may be a 5G core network (referred to as “5GC **120**” or “5G core network **120**”), and the RAN **110** may be connected with the CN **120** using a next generation interface **113**. In some implementations, the next generation interface **113** may be split into two parts, an next generation user plane (NG-U) interface **114**, which carries traffic data between the RAN nodes **111** and a user plane function (UPF), and the S1 control plane (NG-C) interface **115**, which is a signaling interface between the RAN nodes **111** and access and mobility management functions (AMFs). Examples where the CN **120** is a 5G core network are discussed in more detail with regard to FIG. 3.

In some implementations, the CN **120** may be an EPC (referred to as “EPC **120**” or the like), and the RAN **110** may be connected with the CN **120** using an S1 interface **113**. In some implementations, the S1 interface **113** may be split into two parts, an S1 user plane (S1-U) interface **114**, which carries traffic data between the RAN nodes **111** and the serving gateway (S-GW), and the S1-MME interface **115**, which is a signaling interface between the RAN nodes **111** and mobility management entities (MMEs).

In some implementations, some or all of the RAN nodes **111** may be implemented as one or more software entities running on server computers as part of a virtual network, which may be referred to as a cloud RAN (CRAN) or a virtual baseband unit pool (vBBUP). The CRAN or vBBUP may implement a RAN function split, such as a packet data convergence protocol (PDCP) split in which radio resource control (RRC) and PDCP layers are operated by the CRAN/vBBUP and other layer two (e.g., data link layer) protocol entities are operated by individual RAN nodes **111**; a medium access control (MAC)/physical layer (PHY) split in which RRC, PDCP, MAC, and radio link control (RLC) layers are operated by the CRAN/vBBUP and the PHY layer is operated by individual RAN nodes **111**; or a “lower PHY” split in which RRC, PDCP, RLC, and MAC layers and upper portions of the PHY layer are operated by the CRAN/vBBUP and lower portions of the PHY layer are operated by individual RAN nodes **111**. This virtualized framework allows the freed-up processor cores of the RAN nodes **111** to perform, for example, other virtualized applications. In some implementations, an individual RAN node **111** may represent individual gNB distributed units (DUs) that are connected to a gNB central unit (CU) using individual F1 interfaces (not shown in FIG. 1). In some implementations, the gNB-DUs may include one or more remote radio heads or RFEMs (see, e.g., FIG. 4), and the gNB-CU may be operated by a server that is located in the RAN **110** (not shown) or by a server pool in a similar manner as the CRAN/vBBUP. Additionally or alternatively, one or more of the RAN nodes **111** may be next generation eNBs (ng-eNBs), including RAN nodes that provide E-UTRA user plane and control plane protocol terminations toward the UEs **101**, and are connected to a 5G core network (e.g., core network **120**) using a next generation interface.

In vehicle-to-everything (V2X) scenarios, one or more of the RAN nodes **111** may be or act as RSUs. The term “Road Side Unit” or “RSU” refers to any transportation infrastructure entity used for V2X communications. A RSU may be implemented in or by a suitable RAN node or a stationary (or relatively stationary) UE, where a RSU implemented in or by a UE may be referred to as a “UE-type RSU,” a RSU implemented in or by an eNB may be referred to as an “eNB-type RSU,” a RSU implemented in or by a gNB may be referred to as a “gNB-type RSU,” and the like. In some

implementations, an RSU is a computing device coupled with radio frequency circuitry located on a roadside that provides connectivity support to passing vehicle UEs **101** (vUEs **101**). The RSU may also include internal data storage circuitry to store intersection map geometry, traffic statistics, media, as well as applications or other software to sense and control ongoing vehicular and pedestrian traffic. The RSU may operate on the 5.9 GHz Direct Short Range Communications (DSRC) band to provide very low latency communications required for high speed events, such as crash avoidance, traffic warnings, and the like. Additionally or alternatively, the RSU may operate on the cellular V2X band to provide the aforementioned low latency communications, as well as other cellular communications services. Additionally or alternatively, the RSU may operate as a Wi-Fi hotspot (2.4 GHz band) or provide connectivity to one or more cellular networks to provide uplink and downlink communications, or both. The computing device(s) and some or all of the radiofrequency circuitry of the RSU may be packaged in a weatherproof enclosure suitable for outdoor installation, and may include a network interface controller to provide a wired connection (e.g., Ethernet) to a traffic signal controller or a backhaul network, or both.

FIG. 2 illustrates an example architecture of a system **200** including a first CN **220**. In this example, the system **200** may implement the LTE standard such that the CN **220** is an EPC **220** that corresponds with CN **120** of FIG. 1. Additionally, the UE **201** may be the same or similar as the UEs **101** of FIG. 1, and the E-UTRAN **210** may be a RAN that is the same or similar to the RAN **110** of FIG. 1, and which may include RAN nodes **111** discussed previously. The CN **220** may comprise MMEs **221**, an S-GW **222**, a PDN gateway (P-GW) **223**, a high-speed packet access (HSS) function **224**, and a serving GPRS support node (SGSN) **225**.

The MMEs **221** may be similar in function to the control plane of legacy SGSN, and may implement mobility management (MM) functions to keep track of the current location of a UE **201**. The MMEs **221** may perform various mobility management procedures to manage mobility aspects in access such as gateway selection and tracking area list management. Mobility management (also referred to as “EPS MM” or “EMM” in E-UTRAN systems) may refer to all applicable procedures, methods, data storage, and other aspects that are used to maintain knowledge about a present location of the UE **201**, provide user identity confidentiality, or perform other like services to users/subscribers, or combinations of them, among others. Each UE **201** and the MME **221** may include an EMM sublayer, and a mobility management context may be established in the UE **201** and the MME **221** when an attach procedure is successfully completed. The mobility management context may be a data structure or database object that stores mobility management-related information of the UE **201**. The MMEs **221** may be coupled with the HSS **224** using a S6a reference point, coupled with the SGSN **225** using a S3 reference point, and coupled with the S-GW **222** using a S11 reference point.

The SGSN **225** may be a node that serves the UE **201** by tracking the location of an individual UE **201** and performing security functions. In addition, the SGSN **225** may perform Inter-EPC node signaling for mobility between 2G/3G and E-UTRAN 3GPP access networks; PDN and S-GW selection as specified by the MMEs **221**; handling of UE **201** time zone functions as specified by the MMEs **221**; and MME selection for handovers to E-UTRAN 3GPP access network, among other functions. The S3 reference

point between the MMEs **221** and the SGSN **225** may enable user and bearer information exchange for inter-3GPP access network mobility in idle or active states, or both.

The HSS **224** may include a database for network users, including subscription-related information to support the network entities’ handling of communication sessions. The EPC **220** may include one or more HSSs **224** depending on the number of mobile subscribers, on the capacity of the equipment, on the organization of the network, or combinations of them, among other features. For example, the HSS **224** can provide support for routing, roaming, authentication, authorization, naming/addressing resolution, location dependencies, among others. A S6a reference point between the HSS **224** and the MMEs **221** may enable transfer of subscription and authentication data for authenticating or authorizing user access to the EPC **220** between HSS **224** and the MMEs **221**.

The S-GW **222** may terminate the S1 interface **113** (“S1-U” in FIG. 2) toward the RAN **210**, and may route data packets between the RAN **210** and the EPC **220**. In addition, the S-GW **222** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities may include lawful intercept, charging, and some policy enforcement. The S11 reference point between the S-GW **222** and the MMEs **221** may provide a control plane between the MMEs **221** and the S-GW **222**. The S-GW **222** may be coupled with the P-GW **223** using a S5 reference point.

The P-GW **223** may terminate a SGi interface toward a PDN **230**. The P-GW **223** may route data packets between the EPC **220** and external networks such as a network including the application server **130** (sometimes referred to as an “AF”) using an IP communications interface **125** (see, e.g., FIG. 1). In some implementations, the P-GW **223** may be communicatively coupled to an application server (e.g., the application server **130** of FIG. 1 or PDN **230** in FIG. 2) using an IP communications interface **125** (see, e.g., FIG. 1). The S5 reference point between the P-GW **223** and the S-GW **222** may provide user plane tunneling and tunnel management between the P-GW **223** and the S-GW **222**. The S5 reference point may also be used for S-GW **222** relocation due to UE **201** mobility and if the S-GW **222** needs to connect to a non-collocated P-GW **223** for the required PDN connectivity. The P-GW **223** may further include a node for policy enforcement and charging data collection (e.g., PCEF (not shown)). Additionally, the SGi reference point between the P-GW **223** and the packet data network (PDN) **230** may be an operator external public, a private PDN, or an intra operator packet data network, for example, for provision of IMS services. The P-GW **223** may be coupled with a policy control and charging rules function (PCRF) **226** using a Gx reference point.

PCRF **226** is the policy and charging control element of the EPC **220**. In a non-roaming scenario, there may be a single PCRF **226** in the Home Public Land Mobile Network (HPLMN) associated with a UE **201**’s Internet Protocol Connectivity Access Network (IP-CAN) session. In a roaming scenario with local breakout of traffic, there may be two PCRFs associated with a UE **201**’s IP-CAN session, a Home PCRF (H-PCRF) within an HPLMN and a Visited PCRF (V-PCRF) within a Visited Public Land Mobile Network (VPLMN). The PCRF **226** may be communicatively coupled to the application server **230** using the P-GW **223**. The application server **230** may signal the PCRF **226** to indicate a new service flow and select the appropriate quality of service (QoS) and charging parameters. The PCRF **226** may provision this rule into a PCEF (not shown) with the

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appropriate traffic flow template (TFT) and QoS class identifier (QCI), which commences the QoS and charging as specified by the application server 230. The Gx reference point between the PCRF 226 and the P-GW 223 may allow for the transfer of QoS policy and charging rules from the PCRF 226 to PCEF in the P-GW 223. A Rx reference point may reside between the PDN 230 (or “AF 230”) and the PCRF 226.

FIG. 3 illustrates an architecture of a system 300 including a second CN 320. The system 300 is shown to include a UE 301, which may be the same or similar to the UEs 101 and UE 201 discussed previously; a RAN 310, which may be the same or similar to the RAN 110 and RAN 210 discussed previously, and which may include RAN nodes 111 discussed previously; and a data network (DN) 303, which may be, for example, operator services, Internet access or 3rd party services; and a 5GC 320. The 5GC 320 may include an authentication server function (AUSF) 322; an access and mobility management function (AMF) 321; a session management function (SMF) 324; a network exposure function (NEF) 323; a policy control function (PCF) 326; a network repository function (NRF) 325; a unified data management (UDM) function 327; an AF 328; a user plane function (UPF) 302; and a network slice selection function (NSSF) 329.

The UPF 302 may act as an anchor point for intra-RAT and inter-RAT mobility, an external PDU session point of interconnect to DN 303, and a branching point to support multi-homed PDU session. The UPF 302 may also perform packet routing and forwarding, perform packet inspection, enforce the user plane part of policy rules, lawfully intercept packets (QoS collection), perform traffic usage reporting, perform QoS handling for a user plane (e.g., packet filtering, gating, UL/DL rate enforcement), perform Uplink Traffic verification (e.g., SDF to QoS flow mapping), transport level packet marking in the uplink and downlink, and perform downlink packet buffering and downlink data notification triggering. UPF 302 may include an uplink classifier to support routing traffic flows to a data network. The DN 303 may represent various network operator services, Internet access, or third party services. DN 303 may include, or be similar to, application server 130 discussed previously. The UPF 302 may interact with the SMF 324 using a N4 reference point between the SMF 324 and the UPF 302.

The AUSF 322 stores data for authentication of UE 301 and handle authentication-related functionality. The AUSF 322 may facilitate a common authentication framework for various access types. The AUSF 322 may communicate with the AMF 321 using a N12 reference point between the AMF 321 and the AUSF 322, and may communicate with the UDM 327 using a N13 reference point between the UDM 327 and the AUSF 322. Additionally, the AUSF 322 may exhibit a Nausf service-based interface.

The AMF 321 is responsible for registration management (e.g., for registering UE 301), connection management, reachability management, mobility management, and lawful interception of AMF-related events, and access authentication and authorization. The AMF 321 may be a termination point for the N11 reference point between the AMF 321 and the SMF 324. The AMF 321 may provide transport for SM messages between the UE 301 and the SMF 324, and act as a transparent proxy for routing SM messages. AMF 321 may also provide transport for SMS messages between UE 301 and an SMSF (not shown in FIG. 3). AMF 321 may act as security anchor function (SEAF), which may include interaction with the AUSF 322 and the UE 301 to, for example, receive an intermediate key that was established as a result

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of the UE 301 authentication process. Where universal subscriber identity module (USIM) based authentication is used, the AMF 321 may retrieve the security material from the AUSF 322. AMF 321 may also include a security context management (SCM) function, which receives a key from the SEAF to derive access-network specific keys. Furthermore, AMF 321 may be a termination point of a RAN control plane interface, which may include or be a N2 reference point between the RAN 310 and the AMF 321. In some implementations, the AMF 321 may be a termination point of NAS (N1) signaling and perform NAS ciphering and integrity protection.

AMF 321 may also support NAS signaling with a UE 301 over a N3 interworking function (IWF) interface (referred to as the “N3IWF”). The N3IWF may be used to provide access to untrusted entities. The N3IWF may be a termination point for the N2 interface between the RAN 310 and the AMF 321 for the control plane, and may be a termination point for the N3 reference point between the RAN 310 and the UPF 302 for the user plane. As such, the AMF 321 may handle N2 signaling from the SMF 324 and the AMF 321 for PDU sessions and QoS, encapsulate/de-encapsulate packets for IPsec and N3 tunneling, mark N3 user-plane packets in the uplink, and enforce QoS corresponding to N3 packet marking taking into account QoS requirements associated with such marking received over N2. The N3IWF may also relay uplink and downlink control-plane NAS signaling between the UE 301 and AMF 321 using a N1 reference point between the UE 301 and the AMF 321, and relay uplink and downlink user-plane packets between the UE 301 and UPF 302. The N3IWF also provides mechanisms for IPsec tunnel establishment with the UE 301. The AMF 321 may exhibit a Namf service-based interface, and may be a termination point for a N14 reference point between two AMFs 321 and a N17 reference point between the AMF 321 and a 5G equipment identity registry (EIR) (not shown in FIG. 3).

The UE 301 may register with the AMF 321 in order to receive network services. Registration management (RM) is used to register or deregister the UE 301 with the network (e.g., AMF 321), and establish a UE context in the network (e.g., AMF 321). The UE 301 may operate in a RM-REGISTERED state or an RM-DEREGISTERED state. In the RM DEREGISTERED state, the UE 301 is not registered with the network, and the UE context in AMF 321 holds no valid location or routing information for the UE 301 so the UE 301 is not reachable by the AMF 321. In the RM REGISTERED state, the UE 301 is registered with the network, and the UE context in AMF 321 may hold a valid location or routing information for the UE 301 so the UE 301 is reachable by the AMF 321. In the RM-REGISTERED state, the UE 301 may perform mobility Registration Update procedures, perform periodic Registration Update procedures triggered by expiration of the periodic update timer (e.g., to notify the network that the UE 301 is still active), and perform a Registration Update procedure to update UE capability information or to re-negotiate protocol parameters with the network, among others.

The AMF 321 may store one or more RM contexts for the UE 301, where each RM context is associated with a specific access to the network. The RM context may be, for example, a data structure or database object, among others, that indicates or stores a registration state per access type and the periodic update timer. The AMF 321 may also store a 5GC mobility management (MM) context that may be the same or similar to the (E) MM context discussed previously. In some implementations, the AMF 321 may store a coverage

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enhancement mode B Restriction parameter of the UE 301 in an associated MM context or RM context. The AMF 321 may also derive the value, when needed, from the UE's usage setting parameter already stored in the UE context (and/or MM/RM context).

Connection management (CM) may be used to establish and release a signaling connection between the UE 301 and the AMF 321 over the N1 interface. The signaling connection is used to enable NAS signaling exchange between the UE 301 and the CN 320, and includes both the signaling connection between the UE and the AN (e.g., RRC connection or UE-N3IWF connection for non-3GPP access) and the N2 connection for the UE 301 between the AN (e.g., RAN 310) and the AMF 321. In some implementations, the UE 301 may operate in one of two CM modes: CM-IDLE mode or CM-CONNECTED mode. When the UE 301 is operating in the CM-IDLE mode, the UE 301 may have no NAS signaling connection established with the AMF 321 over the N1 interface, and there may be RAN 310 signaling connection (e.g., N2 or N3 connections, or both) for the UE 301. When the UE 301 is operating in the CM-CONNECTED mode, the UE 301 may have an established NAS signaling connection with the AMF 321 over the N1 interface, and there may be a RAN 310 signaling connection (e.g., N2 and/or N3 connections) for the UE 301. Establishment of a N2 connection between the RAN 310 and the AMF 321 may cause the UE 301 to transition from the CM-IDLE mode to the CM-CONNECTED mode, and the UE 301 may transition from the CM-CONNECTED mode to the CM-IDLE mode when N2 signaling between the RAN 310 and the AMF 321 is released.

The SMF 324 may be responsible for session management (SM), such as session establishment, modify and release, including tunnel maintain between UPF and AN node; UE IP address allocation and management (including optional authorization); selection and control of UP function; configuring traffic steering at the UPF to route traffic to proper destination; termination of interfaces toward policy control functions; controlling part of policy enforcement and QoS; lawful intercept (for SM events and interface to LI system); termination of SM parts of NAS messages; down-link data notification; initiating AN specific SM information, sent using AMF over N2 to AN; and determining SSC mode of a session. SM may refer to management of a PDU session, and a PDU session (or "session") may refer to a PDU connectivity service that provides or enables the exchange of PDUs between a UE 301 and a data network (DN) 303 identified by a Data Network Name (DNN). PDU sessions may be established upon UE 301 request, modified upon UE 301 and 5GC 320 request, and released upon UE 301 and 5GC 320 request using NAS SM signaling exchanged over the N1 reference point between the UE 301 and the SMF 324. Upon request from an application server, the 5GC 320 may trigger a specific application in the UE 301. In response to receipt of the trigger message, the UE 301 may pass the trigger message (or relevant parts/information of the trigger message) to one or more identified applications in the UE 301. The identified application(s) in the UE 301 may establish a PDU session to a specific DNN. The SMF 324 may check whether the UE 301 requests are compliant with user subscription information associated with the UE 301. In this regard, the SMF 324 may retrieve and/or request to receive update notifications on SMF 324 level subscription data from the UDM 327.

The SMF 324 may include some or all of the following roaming functionality: handling local enforcement to apply QoS service level agreements (SLAs) (e.g., in VPLMN);

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charging data collection and charging interface (e.g., in VPLMN); lawful intercept (e.g., in VPLMN for SM events and interface to LI system); and support for interaction with external DN for transport of signaling for PDU session authorization/authentication by external DN. A N16 reference point between two SMFs 324 may be included in the system 300, which may be between another SMF 324 in a visited network and the SMF 324 in the home network in roaming scenarios. Additionally, the SMF 324 may exhibit the Nsmf service-based interface.

The NEF 323 may provide means for securely exposing the services and capabilities provided by 3GPP network functions for third party, internal exposure/re-exposure, Application Functions (e.g., AF 328), edge computing or fog computing systems, among others. In some implementations, the NEF 323 may authenticate, authorize, and/or throttle the AFs. The NEF 323 may also translate information exchanged with the AF 328 and information exchanged with internal network functions. For example, the NEF 323 may translate between an AF-Service-Identifier and an internal 5GC information. NEF 323 may also receive information from other network functions (NFs) based on exposed capabilities of other network functions. This information may be stored at the NEF 323 as structured data, or at a data storage NF using standardized interfaces. The stored information can then be re-exposed by the NEF 323 to other NFs and AFs, or used for other purposes such as analytics, or both. Additionally, the NEF 323 may exhibit a Nnef service-based interface.

The NRF 325 may support service discovery functions, receive NF discovery requests from NF instances, and provide the information of the discovered NF instances to the NF instances. NRF 325 also maintains information of available NF instances and their supported services. As used herein, the terms "instantiate," "instantiation," and the like may refer to the creation of an instance, and an "instance" may refer to a concrete occurrence of an object, which may occur, for example, during execution of program code. Additionally, the NRF 325 may exhibit the Nnrf service-based interface.

The PCF 326 may provide policy rules to control plane function(s) to enforce them, and may also support unified policy framework to govern network behavior. The PCF 326 may also implement a front end to access subscription information relevant for policy decisions in a unified data repository (UDR) of the UDM 327. The PCF 326 may communicate with the AMF 321 using an N15 reference point between the PCF 326 and the AMF 321, which may include a PCF 326 in a visited network and the AMF 321 in case of roaming scenarios. The PCF 326 may communicate with the AF 328 using a N5 reference point between the PCF 326 and the AF 328; and with the SMF 324 using a N7 reference point between the PCF 326 and the SMF 324. The system 300 or CN 320, or both, may also include a N24 reference point between the PCF 326 (in the home network) and a PCF 326 in a visited network. Additionally, the PCF 326 may exhibit a Npcf service-based interface.

The UDM 327 may handle subscription-related information to support the network entities' handling of communication sessions, and may store subscription data of UE 301. For example, subscription data may be communicated between the UDM 327 and the AMF 321 using a N8 reference point between the UDM 327 and the AMF. The UDM 327 may include two parts, an application front end and a UDR (the front end and UDR are not shown in FIG. 3). The UDR may store subscription data and policy data for the UDM 327 and the PCF 326, or structured data for

exposure and application data (including PFDs for application detection, application request information for multiple UEs 301) for the NEF 323, or both. The Nudr service-based interface may be exhibited by the UDR 221 to allow the UDM 327, PCF 326, and NEF 323 to access a particular set of the stored data, as well as to read, update (e.g., add, modify), delete, and subscribe to notification of relevant data changes in the UDR. The UDM may include a UDM front end, which is in charge of processing credentials, location management, subscription management and so on. Several different front ends may serve the same user in different transactions. The UDM front end accesses subscription information stored in the UDR and performs authentication credential processing, user identification handling, access authorization, registration/mobility management, and subscription management. The UDR may interact with the SMF 324 using a N10 reference point between the UDM 327 and the SMF 324. UDM 327 may also support SMS management, in which an SMS front end implements the similar application logic as discussed previously. Additionally, the UDM 327 may exhibit the Nudm service-based interface.

The AF 328 may provide application influence on traffic routing, provide access to the network capability exposure (NCE), and interact with the policy framework for policy control. The NCE may be a mechanism that allows the 5GC 320 and AF 328 to provide information to each other using NEF 323, which may be used for edge computing implementations. In such implementations, the network operator and third party services may be hosted close to the UE 301 access point of attachment to achieve an efficient service delivery through the reduced end-to-end latency and load on the transport network. For edge computing implementations, the 5GC may select a UPF 302 close to the UE 301 and execute traffic steering from the UPF 302 to DN 303 using the N6 interface. This may be based on the UE subscription data, UE location, and information provided by the AF 328. In this way, the AF 328 may influence UPF (re) selection and traffic routing. Based on operator deployment, when AF 328 is considered to be a trusted entity, the network operator may permit AF 328 to interact directly with relevant NFs. Additionally, the AF 328 may exhibit a Naf service-based interface.

The NSSF 329 may select a set of network slice instances serving the UE 301. The NSSF 329 may also determine allowed NSSAI and the mapping to the subscribed single network slice selection assistance information (S-NSSAI), if needed. The NSSF 329 may also determine the AMF set to be used to serve the UE 301, or a list of candidate AMF(s) 321 based on a suitable configuration and possibly by querying the NRF 325. The selection of a set of network slice instances for the UE 301 may be triggered by the AMF 321 with which the UE 301 is registered by interacting with the NSSF 329, which may lead to a change of AMF 321. The NSSF 329 may interact with the AMF 321 using an N22 reference point between AMF 321 and NSSF 329; and may communicate with another NSSF 329 in a visited network using a N31 reference point (not shown by FIG. 3). Additionally, the NSSF 329 may exhibit a Nnssf service-based interface.

As discussed previously, the CN 320 may include an SMSF, which may be responsible for SMS subscription checking and verification, and relaying SM messages to or from the UE 301 to or from other entities, such as an SMS-GMSC/IW MSC/SMS-router. The SMS may also interact with AMF 321 and UDM 327 for a notification procedure that the UE 301 is available for SMS transfer

(e.g., set a UE not reachable flag, and notifying UDM 327 when UE 301 is available for SMS).

In some implementations, there may be additional or alternative reference points or service-based interfaces, or both, between the network function services in the network functions. However, these interfaces and reference points have been omitted from FIG. 3 for clarity. In one example, the CN 320 may include a Nx interface, which is an inter-CN interface between the MME (e.g., MME 221) and the AMF 321 in order to enable interworking between CN 320 and CN 220. Other example interfaces or reference points may include a N5g-EIR service-based interface exhibited by a 5G-EIR, a N27 reference point between the NRF in the visited network and the NRF in the home network, or a N31 reference point between the NSSF in the visited network and the NSSF in the home network, among others.

In some implementations, the components of the CN 220 may be implemented in one physical node or separate physical nodes and may include components to read and execute instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium). In some implementations, the components of CN 320 may be implemented in a same or similar manner as discussed herein with regard to the components of CN 220. In some implementations, NFV is utilized to virtualize any or all of the above-described network node functions using executable instructions stored in one or more computer-readable storage mediums, as described in further detail below. A logical instantiation of the CN 220 may be referred to as a network slice, and individual logical instantiations of the CN 220 may provide specific network capabilities and network characteristics. A logical instantiation of a portion of the CN 220 may be referred to as a network sub-slice, which can include the P-GW 223 and the PCRF 226.

As used herein, the terms “instantiate,” “instantiation,” and the like may refer to the creation of an instance, and an “instance” may refer to a concrete occurrence of an object, which may occur, for example, during execution of program code. A network instance may refer to information identifying a domain, which may be used for traffic detection and routing in case of different IP domains or overlapping IP addresses. A network slice instance may refer to a set of network functions (NFs) instances and the resources (e.g., compute, storage, and networking resources) required to deploy the network slice.

With respect to 5G systems (see, e.g., FIG. 3), a network slice may include a RAN part and a CN part. The support of network slicing relies on the principle that traffic for different slices is handled by different PDU sessions. The network can realize the different network slices by scheduling or by providing different L1/L2 configurations, or both. The UE 301 provides assistance information for network slice selection in an appropriate RRC message if it has been provided by NAS. While the network can support large number of slices, the UE need not support more than 8 slices simultaneously in some implementations.

A network slice may include the CN 320 control plane and user plane NFs, NG-RANs 310 in a serving PLMN, and a N3IWF functions in the serving PLMN. Individual network slices may have different S-NSSAI or different SSTs, or both. NSSAI includes one or more S-NSSAIs, and each network slice is uniquely identified by an S-NSSAI. Network slices may differ for supported features and network functions optimizations. In some implementations, multiple network slice instances may deliver the same services or features but for different groups of UEs 301 (e.g., enterprise

users). For example, individual network slices may deliver different committed service(s) or may be dedicated to a particular customer or enterprise, or both. In this example, each network slice may have different S-NSSAIs with the same SST but with different slice differentiators. Additionally, a single UE may be served with one or more network slice instances simultaneously using a 5G AN, and the UE may be associated with eight different S-NSSAIs. Moreover, an AMF 321 instance serving an individual UE 301 may belong to each of the network slice instances serving that UE.

Network slicing in the NG-RAN 310 involves RAN slice awareness. RAN slice awareness includes differentiated handling of traffic for different network slices, which have been pre-configured. Slice awareness in the NG-RAN 310 is introduced at the PDU session level by indicating the S-NSSAI corresponding to a PDU session in all signaling that includes PDU session resource information. How the NG-RAN 310 supports the slice enabling in terms of NG-RAN functions (e.g., the set of network functions that comprise each slice) is implementation dependent. The NG-RAN 310 selects the RAN part of the network slice using assistance information provided by the UE 301 or the 5GC 320, which unambiguously identifies one or more of the pre-configured network slices in the PLMN. The NG-RAN 310 also supports resource management and policy enforcement between slices as per SLAs. A single NG-RAN node may support multiple slices, and the NG-RAN 310 may also apply an appropriate RRM policy for the SLA in place to each supported slice. The NG-RAN 310 may also support QoS differentiation within a slice.

The NG-RAN 310 may also use the UE assistance information for the selection of an AMF 321 during an initial attach, if available. The NG-RAN 310 uses the assistance information for routing the initial NAS to an AMF 321. If the NG-RAN 310 is unable to select an AMF 321 using the assistance information, or the UE 301 does not provide any such information, the NG-RAN 310 sends the NAS signaling to a default AMF 321, which may be among a pool of AMFs 321. For subsequent accesses, the UE 301 provides a temp ID, which is assigned to the UE 301 by the 5GC 320, to enable the NG-RAN 310 to route the NAS message to the appropriate AMF 321 as long as the temp ID is valid. The NG-RAN 310 is aware of, and can reach, the AMF 321 that is associated with the temp ID. Otherwise, the method for initial attach applies.

The NG-RAN 310 supports resource isolation between slices. NG-RAN 310 resource isolation may be achieved by means of RRM policies and protection mechanisms that should avoid that shortage of shared resources if one slice breaks the service level agreement for another slice. In some implementations, it is possible to fully dedicate NG-RAN 310 resources to a certain slice. How NG-RAN 310 supports resource isolation is implementation dependent.

Some slices may be available only in part of the network. Awareness in the NG-RAN 310 of the slices supported in the cells of its neighbors may be beneficial for inter-frequency mobility in connected mode. The slice availability may not change within the UE's registration area. The NG-RAN 310 and the 5GC 320 are responsible to handle a service request for a slice that may or may not be available in a given area. Admission or rejection of access to a slice may depend on factors such as support for the slice, availability of resources, support of the requested service by NG-RAN 310.

The UE 301 may be associated with multiple network slices simultaneously. In case the UE 301 is associated with multiple slices simultaneously, only one signaling connec-

tion is maintained, and for intra-frequency cell reselection, the UE 301 tries to camp on the best cell. For inter-frequency cell reselection, dedicated priorities can be used to control the frequency on which the UE 301 camps. The 5GC 320 is to validate that the UE 301 has the rights to access a network slice. Prior to receiving an Initial Context Setup Request message, the NG-RAN 310 may be allowed to apply some provisional or local policies based on awareness of a particular slice that the UE 301 is requesting to access. During the initial context setup, the NG-RAN 310 is informed of the slice for which resources are being requested.

FIG. 4 illustrates an example of infrastructure equipment 400. The infrastructure equipment 400 (or "system 400") may be implemented as a base station, a radio head, a RAN node, such as the RAN nodes 111 shown and described previously, an application server 130, or any other component or device described herein. In other examples, the system 400 can be implemented in or by a UE.

The system 400 includes application circuitry 405, baseband circuitry 410, one or more radio front end modules (RFEMs) 415, memory circuitry 420, power management integrated circuitry (PMIC) 425, power tee circuitry 430, network controller circuitry 435, network interface connector 440, satellite positioning circuitry 445, and user interface circuitry 450. In some implementations, the system 400 may include additional elements such as, for example, memory, storage, a display, a camera, one or more sensors, or an input/output (I/O) interface, or combinations of them, among others. In other examples, the components described with reference to the system 400 may be included in more than one device. For example, the various circuitries may be separately included in more than one device for CRAN, vBBU, or other implementations.

The application circuitry 405 includes circuitry such as, but not limited to, one or more processors (or processor cores), cache memory, one or more of low drop-out voltage regulators (LDOs), interrupt controllers, serial interfaces such as SPI, I2C or universal programmable serial interface module, real time clock (RTC), timer-counters including interval and watchdog timers, general purpose input/output (I/O or IO), memory card controllers such as Secure Digital (SD), MultiMediaCard (MMC), Universal Serial Bus (USB) interfaces, Mobile Industry Processor Interface (MIPI) interfaces and Joint Test Access Group (JTAG) test access ports. The processors (or cores) of the application circuitry 405 may be coupled with or may include memory or storage elements and may be configured to execute instructions stored in the memory or storage to enable various applications or operating systems to run on the system 400. In some implementations, the memory or storage elements may include on-chip memory circuitry, which may include any suitable volatile or non-volatile memory, such as DRAM, SRAM, EPROM, EEPROM, Flash memory, solid-state memory, or combinations of them, among other types of memory.

The processor(s) of the application circuitry 405 may include, for example, one or more processor cores (CPUs), one or more application processors, one or more graphics processing units (GPUs), one or more reduced instruction set computing (RISC) processors, one or more complex instruction set computing (CISC) processors, one or more digital signal processors (DSP), one or more FPGAs, one or more PLDs, one or more ASICs, one or more microprocessors or controllers, or combinations of them, among others. In some implementations, the application circuitry 405 may include, or may be, a special-purpose processor or controller configured to carry out the various techniques described

here. In some implementations, the system **400** may not utilize application circuitry **405**, and instead may include a special-purpose processor or controller to process IP data received from an EPC or 5GC, for example.

In some implementations, the application circuitry **405** may include one or more hardware accelerators, which may be microprocessors, programmable processing devices, or the like. The one or more hardware accelerators may include, for example, computer vision (CV) or deep learning (DL) accelerators, or both. In some implementations, the programmable processing devices may be one or more a field-programmable devices (FPDs) such as field-programmable gate arrays (FPGAs) and the like; programmable logic devices (PLDs) such as complex PLDs (CPLDs) or high-capacity PLDs (HCPLDs); ASICs such as structured ASICs; programmable SoCs (PSoCs), or combinations of them, among others. In such implementations, the circuitry of application circuitry **405** may include logic blocks or logic fabric, and other interconnected resources that may be programmed to perform various functions, such as the procedures, methods, functions described herein. In some implementations, the circuitry of application circuitry **405** may include memory cells (e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash memory, static memory (e.g., static random access memory (SRAM) or anti-fuses)) used to store logic blocks, logic fabric, data, or other data in look-up-tables (LUTs) and the like.

The user interface circuitry **450** may include one or more user interfaces designed to enable user interaction with the system **400** or peripheral component interfaces designed to enable peripheral component interaction with the system **400**. User interfaces may include, but are not limited to, one or more physical or virtual buttons (e.g., a reset button), one or more indicators (e.g., light emitting diodes (LEDs)), a physical keyboard or keypad, a mouse, a touchpad, a touch-screen, speakers or other audio emitting devices, microphones, a printer, a scanner, a headset, a display screen or display device, or combinations of them, among others. Peripheral component interfaces may include, but are not limited to, a nonvolatile memory port, a universal serial bus (USB) port, an audio jack, a power supply interface, among others.

The radio front end modules (RFEMs) **415** may include a millimeter wave (mmWave) RFEM and one or more sub-mmWave radio frequency integrated circuits (RFICs). In some implementations, the one or more sub-mmWave RFICs may be physically separated from the mmWave RFEM. The RFICs may include connections to one or more antennas or antenna arrays (see, e.g., antenna array **611** of FIG. **6**), and the RFEM may be connected to multiple antennas. In some implementations, both mmWave and sub-mmWave radio functions may be implemented in the same physical RFEM **415**, which incorporates both mmWave antennas and sub-mm Wave. The baseband circuitry **410** may be implemented, for example, as a solder-down substrate including one or more integrated circuits, a single packaged integrated circuit soldered to a main circuit board or a multi-chip module containing two or more integrated circuits.

The memory circuitry **420** may include one or more of volatile memory, such as dynamic random access memory (DRAM) or synchronous dynamic random access memory (SDRAM), and nonvolatile memory (NVM), such as high-speed electrically erasable memory (commonly referred to as Flash memory), phase change random access memory (PRAM), or magnetoresistive random access memory

(MRAM), or combinations of them, among others. Memory circuitry **420** may be implemented as one or more of solder down packaged integrated circuits, socketed memory modules and plug-in memory cards, for example.

The PMIC **425** may include voltage regulators, surge protectors, power alarm detection circuitry, and one or more backup power sources such as a battery or capacitor. The power alarm detection circuitry may detect one or more of brown out (under-voltage) and surge (over-voltage) conditions. The power tee circuitry **430** may provide for electrical power drawn from a network cable to provide both power supply and data connectivity to the infrastructure equipment **400** using a single cable.

The network controller circuitry **435** may provide connectivity to a network using a standard network interface protocol such as Ethernet, Ethernet over GRE Tunnels, Ethernet over Multiprotocol Label Switching (MPLS), or some other suitable protocol. Network connectivity may be provided to and from the infrastructure equipment **400** using network interface connector **440** using a physical connection, which may be electrical (commonly referred to as a "copper interconnect"), optical, or wireless. The network controller circuitry **435** may include one or more dedicated processors or FPGAs, or both, to communicate using one or more of the aforementioned protocols. In some implementations, the network controller circuitry **435** may include multiple controllers to provide connectivity to other networks using the same or different protocols.

The positioning circuitry **445** includes circuitry to receive and decode signals transmitted or broadcasted by a positioning network of a global navigation satellite system (GNSS). Examples of a GNSS include United States' Global Positioning System (GPS), Russia's Global Navigation System (GLONASS), the European Union's Galileo system, China's BeiDou Navigation Satellite System, a regional navigation system or GNSS augmentation system (e.g., Navigation with Indian Constellation (NAVIC), Japan's Quasi-Zenith Satellite System (QZSS), France's Doppler Orbitography and Radio-positioning Integrated by Satellite (DORIS)), among other systems. The positioning circuitry **445** can include various hardware elements (e.g., including hardware devices such as switches, filters, amplifiers, antenna elements, and the like to facilitate OTA communications) to communicate with components of a positioning network, such as navigation satellite constellation nodes. In some implementations, the positioning circuitry **445** may include a Micro-Technology for Positioning, Navigation, and Timing (Micro-PNT) IC that uses a master timing clock to perform position tracking and estimation without GNSS assistance. The positioning circuitry **445** may also be part of, or interact with, the baseband circuitry **410** or RFEMs **415**, or both, to communicate with the nodes and components of the positioning network. The positioning circuitry **445** may also provide data (e.g., position data, time data) to the application circuitry **405**, which may use the data to synchronize operations with various infrastructure (e.g., RAN nodes **111**).

FIG. **5** illustrates an example of a platform **500** (or "device **500**"). In some implementations, the computer platform **500** may be suitable for use as UEs **101**, **201**, **301**, application servers **130**, or any other component or device discussed herein. The platform **500** may include any combinations of the components shown in the example. The components of platform **500** (or portions thereof) may be implemented as integrated circuits (ICs), discrete electronic devices, or other modules, logic, hardware, software, firmware, or a combination of them adapted in the computer

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platform **500**, or as components otherwise incorporated within a chassis of a larger system. The block diagram of FIG. **5** is intended to show a high level view of components of the platform **500**. However, in some implementations, the platform **500** may include fewer, additional, or alternative components, or a different arrangement of the components shown in FIG. **5**.

The application circuitry **505** includes circuitry such as, but not limited to, one or more processors (or processor cores), cache memory, and one or more of LDOs, interrupt controllers, serial interfaces such as SPI, I2C or universal programmable serial interface module, RTC, timer-counters including interval and watchdog timers, general purpose I/O, memory card controllers such as SD MMC or similar, USB interfaces, MIPI interfaces, and JTAG test access ports. The processors (or cores) of the application circuitry **505** may be coupled with or may include memory/storage elements and may be configured to execute instructions stored in the memory or storage to enable various applications or operating systems to run on the system **500**. In some implementations, the memory or storage elements may be on-chip memory circuitry, which may include any suitable volatile or non-volatile memory, such as DRAM, SRAM, EPROM, EEPROM, Flash memory, solid-state memory, or combinations of them, among other types of memory.

The processor(s) of application circuitry **505** may include, for example, one or more processor cores, one or more application processors, one or more GPUs, one or more RISC processors, one or more ARM processors, one or more CISC processors, one or more DSP, one or more FPGAs, one or more PLDs, one or more ASICs, one or more microprocessors or controllers, a multithreaded processor, an ultra-low voltage processor, an embedded processor, some other known processing element, or any suitable combination thereof. In some implementations, the application circuitry **405** may include, or may be, a special-purpose processor/controller to carry out the techniques described herein. In some implementations, the application circuitry **505** may be a part of a system on a chip (SoC) in which the application circuitry **505** and other components are formed into a single integrated circuit, or a single package.

In some implementations, the application circuitry **505** may include circuitry such as, but not limited to, one or more a field-programmable devices (FPDs) such as FPGAs; PLDs such as CPLDs, HCPLDs; ASICs such as structured ASICs; PSoCs, or combinations of them, among others. In some implementations, the application circuitry **505** may include logic blocks or logic fabric, and other interconnected resources that may be programmed to perform various functions, such as the procedures, methods, functions described herein. In some implementations, the application circuitry **505** may include memory cells e.g., EPROM, EEPROM, flash memory, static memory such as SRAM or anti-fuses, used to store logic blocks, logic fabric, data, or other data in LUTs and the like.

The baseband circuitry **510** may be implemented, for example, as a solder-down substrate including one or more integrated circuits, a single packaged integrated circuit soldered to a main circuit board or a multi-chip module containing two or more integrated circuits. The various hardware electronic elements of baseband circuitry **510** are discussed with regard to FIG. **6**.

The RFEMs **515** may comprise a millimeter wave (mm-Wave) RFEM and one or more sub-mmWave RFICs. In some implementations, the one or more sub-mmWave RFICs may be physically separated from the mmWave RFEM. The RFICs may include connections to one or more

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antennas or antenna arrays (see, e.g., antenna array **611** of FIG. **6**), and the RFEM may be connected to multiple antennas. In some implementations, both mmWave and sub-mmWave radio functions may be implemented in the same physical RFEM **515**, which incorporates both mmWave antennas and sub-mmWave. In some implementations, the RFEMs **515**, the baseband circuitry **510**, or both are included in a transceiver of the platform **500**.

The memory circuitry **520** may include any number and type of memory devices used to provide for a given amount of system memory. As examples, the memory circuitry **520** may include one or more of volatile memory, such as RAM, DRAM, or SDRAM, and NVM, such as high-speed electrically erasable memory (commonly referred to as Flash memory), PRAM, or MRAM, or combinations of them, among others. In low power implementations, the memory circuitry **520** may be on-die memory or registers associated with the application circuitry **505**. To provide for persistent storage of information such as data, applications, operating systems and so forth, memory circuitry **520** may include one or more mass storage devices, which may include, for example, a solid state drive (SSD), hard disk drive (HDD), a micro HDD, resistance change memories, phase change memories, holographic memories, or chemical memories, among others.

The removable memory circuitry **523** may include devices, circuitry, enclosures, housings, ports or receptacles, among others, used to couple portable data storage devices with the platform **500**. These portable data storage devices may be used for mass storage purposes, and may include, for example, flash memory cards (e.g., Secure Digital (SD) cards, microSD cards, XD picture cards), and USB flash drives, optical discs, or external HDDs, or combinations of them, among others. The platform **500** may also include interface circuitry (not shown) for connecting external devices with the platform **500**. The external devices connected to the platform **500** using the interface circuitry include sensor circuitry **521** and electro-mechanical components (EMCs) **522**, as well as removable memory devices coupled to removable memory circuitry **523**.

The sensor circuitry **521** include devices, modules, or subsystems whose purpose is to detect events or changes in its environment and send the information (e.g., sensor data) about the detected events to one or more other devices, modules, or subsystems. Examples of such sensors include inertial measurement units (IMUs) such as accelerometers, gyroscopes, or magnetometers; microelectromechanical systems (MEMS) or nanoelectromechanical systems (NEMS) including 3-axis accelerometers, 3-axis gyroscopes, or magnetometers; level sensors; flow sensors; temperature sensors (e.g., thermistors); pressure sensors; barometric pressure sensors; gravimeters; altimeters; image capture devices (e.g., cameras or lensless apertures); light detection and ranging (LiDAR) sensors; proximity sensors (e.g., infrared radiation detector and the like), depth sensors, ambient light sensors, ultrasonic transceivers; microphones or other audio capture devices, or combinations of them, among others.

The EMCs **522** include devices, modules, or subsystems whose purpose is to enable the platform **500** to change its state, position, or orientation, or move or control a mechanism, system, or subsystem. Additionally, the EMCs **522** may be configured to generate and send messages or signaling to other components of the platform **500** to indicate a current state of the EMCs **522**. Examples of the EMCs **522** include one or more power switches, relays, such as electromechanical relays (EMRs) or solid state relays (SSRs), actuators (e.g., valve actuators), an audible sound generator,

a visual warning device, motors (e.g., DC motors or stepper motors), wheels, thrusters, propellers, claws, clamps, hooks, or combinations of them, among other electro-mechanical components. In some implementations, the platform 500 is configured to operate one or more EMCs 522 based on one or more captured events, instructions, or control signals received from a service provider or clients, or both.

In some implementations, the interface circuitry may connect the platform 500 with positioning circuitry 545. The positioning circuitry 545 includes circuitry to receive and decode signals transmitted or broadcasted by a positioning network of a GNSS. The positioning circuitry 545 comprises various hardware elements (e.g., including hardware devices such as switches, filters, amplifiers, antenna elements, and the like to facilitate OTA communications) to communicate with components of a positioning network, such as navigation satellite constellation nodes. In some implementations, the positioning circuitry 545 may include a Micro-PNT IC that uses a master timing clock to perform position tracking or estimation without GNSS assistance. The positioning circuitry 545 may also be part of, or interact with, the baseband circuitry 510 or RFEMs 515, or both, to communicate with the nodes and components of the positioning network. The positioning circuitry 545 may also provide data (e.g., position data, time data) to the application circuitry 505, which may use the data to synchronize operations with various infrastructure (e.g., radio base stations), for turn-by-turn navigation applications, or the like.

In some implementations, the interface circuitry may connect the platform 500 with Near-Field Communication (NFC) circuitry 540. The NFC circuitry 540 is configured to provide contactless, short-range communications based on radio frequency identification (RFID) standards, in which magnetic field induction is used to enable communication between NFC circuitry 540 and NFC-enabled devices external to the platform 500 (e.g., an “NFC touchpoint”). The NFC circuitry 540 includes an NFC controller coupled with an antenna element and a processor coupled with the NFC controller. The NFC controller may be a chip or IC providing NFC functionalities to the NFC circuitry 540 by executing NFC controller firmware and an NFC stack. The NFC stack may be executed by the processor to control the NFC controller, and the NFC controller firmware may be executed by the NFC controller to control the antenna element to emit short-range RF signals. The RF signals may power a passive NFC tag (e.g., a microchip embedded in a sticker or wristband) to transmit stored data to the NFC circuitry 540, or initiate data transfer between the NFC circuitry 540 and another active NFC device (e.g., a smartphone or an NFC-enabled POS terminal) that is proximate to the platform 500.

The driver circuitry 546 may include software and hardware elements that operate to control particular devices that are embedded in the platform 500, attached to the platform 500, or otherwise communicatively coupled with the platform 500. The driver circuitry 546 may include individual drivers allowing other components of the platform 500 to interact with or control various input/output (I/O) devices that may be present within, or connected to, the platform 500. For example, the driver circuitry 546 may include a display driver to control and allow access to a display device, a touchscreen driver to control and allow access to a touchscreen interface of the platform 500, sensor drivers to obtain sensor readings of sensor circuitry 521 and control and allow access to sensor circuitry 521, EMC drivers to obtain actuator positions of the EMCs 522 or control and allow access to the EMCs 522, a camera driver to control

and allow access to an embedded image capture device, audio drivers to control and allow access to one or more audio devices.

The power management integrated circuitry (PMIC) 525 (also referred to as “power management circuitry 525”) may manage power provided to various components of the platform 500. In particular, with respect to the baseband circuitry 510, the PMIC 525 may control power-source selection, voltage scaling, battery charging, or DC-to-DC conversion. The PMIC 525 may be included when the platform 500 is capable of being powered by a battery 530, for example, when the device is included in a UE 101, 201, 301.

In some implementations, the PMIC 525 may control, or otherwise be part of, various power saving mechanisms of the platform 500. For example, if the platform 500 is in an RRC_CONNECTED state, where it is still connected to the RAN node as it expects to receive traffic shortly, then it may enter a state known as Discontinuous Reception Mode (DRX) after a period of inactivity. During this state, the platform 500 may power down for brief intervals of time and thus save power. If there is no data traffic activity for an extended period of time, then the platform 500 may transition off to an RRC_IDLE state, where it disconnects from the network and does not perform operations such as channel quality feedback or handover. This can allow the platform 500 to enter a very low power state, where it periodically wakes up to listen to the network and then powers down again. In some implementations, the platform 500 may not receive data in the RRC_IDLE state and instead must transition back to RRC_CONNECTED state to receive data. An additional power saving mode may allow a device to be unavailable to the network for periods longer than a paging interval (ranging from seconds to a few hours). During this time, the device may be unreachable to the network and may power down completely. Any data sent during this time may incur a large delay and it is assumed the delay is acceptable.

A battery 530 may power the platform 500, although in some implementations the platform 500 may be deployed in a fixed location, and may have a power supply coupled to an electrical grid. The battery 530 may be a lithium ion battery, a metal-air battery, such as a zinc-air battery, an aluminum-air battery, or a lithium-air battery, among others. In some implementations, such as in V2X applications, the battery 530 may be a typical lead-acid automotive battery.

In some implementations, the battery 530 may be a “smart battery,” which includes or is coupled with a Battery Management System (BMS) or battery monitoring integrated circuitry. The BMS may be included in the platform 500 to track the state of charge (SoCh) of the battery 530. The BMS may be used to monitor other parameters of the battery 530 to provide failure predictions, such as the state of health (SoH) and the state of function (SoF) of the battery 530. The BMS may communicate the information of the battery 530 to the application circuitry 505 or other components of the platform 500. The BMS may also include an analog-to-digital (ADC) convertor that allows the application circuitry 505 to directly monitor the voltage of the battery 530 or the current flow from the battery 530. The battery parameters may be used to determine actions that the platform 500 may perform, such as transmission frequency, network operation, or sensing frequency, among others.

The user interface circuitry 550 includes various input/output (I/O) devices present within, or connected to, the platform 500, and includes one or more user interfaces designed to enable user interaction with the platform 500 or peripheral component interfaces designed to enable periph-

eral component interaction with the platform **500**. The user interface circuitry **550** includes input device circuitry and output device circuitry. Input device circuitry includes any physical or virtual means for accepting an input including one or more physical or virtual buttons (e.g., a reset button), a physical keyboard, keypad, mouse, touchpad, touchscreen, microphones, scanner, or headset, or combinations of them, among others. The output device circuitry includes any physical or virtual means for showing information or otherwise conveying information, such as sensor readings, actuator position(s), or other information. Output device circuitry may include any number or combinations of audio or visual display, including one or more simple visual outputs or indicators (e.g., binary status indicators (e.g., light emitting diodes (LEDs)), multi-character visual outputs, or more complex outputs such as display devices or touchscreens (e.g., Liquid Crystal Displays (LCD), LED displays, quantum dot displays, or projectors), with the output of characters, graphics, or multimedia objects being generated or produced from the operation of the platform **500**. The output device circuitry may also include speakers or other audio emitting devices, or printer(s). In some implementations, the sensor circuitry **521** may be used as the input device circuitry (e.g., an image capture device or motion capture device), and one or more EMCs may be used as the output device circuitry (e.g., an actuator to provide haptic feedback). In another example, NFC circuitry comprising an NFC controller coupled with an antenna element and a processing device may be included to read electronic tags or connect with another NFC-enabled device. Peripheral component interfaces may include, but are not limited to, a non-volatile memory port, a USB port, an audio jack, or a power supply interface.

FIG. 6 illustrates example components of baseband circuitry **610** and radio front end modules (RFEM) **615**. The baseband circuitry **610** can correspond to the baseband circuitry **410** and **510** of FIGS. 4 and 5, respectively. The RFEM **615** can correspond to the RFEM **415** and **515** of FIGS. 4 and 5, respectively. As shown, the RFEMs **615** may include Radio Frequency (RF) circuitry **606**, front-end module (FEM) circuitry **608**, and antenna array **611** coupled together. In some implementations, the RFEMs **615**, the baseband circuitry **610**, or both are included in a transceiver.

The baseband circuitry **610** includes circuitry configured to carry out various radio or network protocol and control functions that enable communication with one or more radio networks using the RF circuitry **606**. The radio control functions may include, but are not limited to, signal modulation and demodulation, encoding and decoding, and radio frequency shifting. In some implementations, modulation and demodulation circuitry of the baseband circuitry **610** may include Fast-Fourier Transform (FFT), precoding, or constellation mapping and demapping functionality. In some implementations, encoding and decoding circuitry of the baseband circuitry **610** may include convolution, tail-biting convolution, turbo, Viterbi, or Low Density Parity Check (LDPC) encoder and decoder functionality. Modulation and demodulation and encoder and decoder functionality are not limited to these examples and may include other suitable functionality in other examples. The baseband circuitry **610** is configured to process baseband signals received from a receive signal path of the RF circuitry **606** and to generate baseband signals for a transmit signal path of the RF circuitry **606**. The baseband circuitry **610** is configured to interface with application circuitry (e.g., the application

operations of the RF circuitry **606**. The baseband circuitry **610** may handle various radio control functions.

The aforementioned circuitry and control logic of the baseband circuitry **610** may include one or more single or multi-core processors. For example, the one or more processors may include a 3G baseband processor **604A**, a 4G or LTE baseband processor **604B**, a 5G or NR baseband processor **604C**, or some other baseband processor(s) **604D** for other existing generations, generations in development or to be developed in the future (e.g., sixth generation (6G)). In some implementations, some or all of the functionality of baseband processors **604A-D** may be included in modules stored in the memory **604G** and executed using one or more processors such as a Central Processing Unit (CPU) **604E**. In some implementations, some or all of the functionality of baseband processors **604A-D** may be provided as hardware accelerators (e.g., FPGAs or ASICs) loaded with the appropriate bit streams or logic blocks stored in respective memory cells. In some implementations, the memory **604G** may store program code of a real-time OS (RTOS) which, when executed by the CPU **604E** (or other processor), is to cause the CPU **604E** (or other processor) to manage resources of the baseband circuitry **610**, schedule tasks, or carry out other operations. In some implementations, the baseband circuitry **610** includes one or more audio digital signal processors (DSP) **604F**. An audio DSP **604F** can include elements for compression and decompression and echo cancellation and may include other suitable processing elements in some implementations.

In some implementations, each of the processors **604A-604E** includes respective memory interfaces to send and receive data to and from the memory **604G**. The baseband circuitry **610** may further include one or more interfaces to communicatively couple to other circuitries or devices, such as an interface to send and receive data to and from memory external to the baseband circuitry **610**; an application circuitry interface to send and receive data to and from the application circuitry **405**, **505** of FIGS. 4 and 5; an RF circuitry interface to send and receive data to and from RF circuitry **606** of FIG. 6; a wireless hardware connectivity interface to send and receive data to and from one or more wireless hardware elements (e.g., Near Field Communication (NFC) components, Bluetooth®/Bluetooth® Low Energy components, Wi-Fi components, and/or the like); and a power management interface to send and receive power or control signals to and from the PMIC **525**.

In some implementations (which may be combined with the above described examples), the baseband circuitry **610** includes one or more digital baseband systems, which are coupled with one another using an interconnect subsystem and to a CPU subsystem, an audio subsystem, and an interface subsystem. The digital baseband subsystems may also be coupled to a digital baseband interface and a mixed-signal baseband subsystem using another interconnect subsystem. Each of the interconnect subsystems may include a bus system, point-to-point connections, network-on-chip (NOC) structures, or some other suitable bus or interconnect technology, such as those discussed herein. The audio subsystem may include DSP circuitry, buffer memory, program memory, speech processing accelerator circuitry, data converter circuitry such as analog-to-digital and digital-to-analog converter circuitry, analog circuitry including one or more of amplifiers and filters, among other components. In some implementations, the baseband circuitry **610** may include protocol processing circuitry with one or more instances of control circuitry (not shown) to provide control

functions for the digital baseband circuitry or radio frequency circuitry (e.g., the radio front end modules **615**).

In some implementations, the baseband circuitry **610** includes individual processing device(s) to operate one or more wireless communication protocols (e.g., a “multi-protocol baseband processor” or “protocol processing circuitry”) and individual processing device(s) to implement PHY layer functions. In some implementations, the PHY layer functions include the aforementioned radio control functions. In some implementations, the protocol processing circuitry operates or implements various protocol layers or entities of one or more wireless communication protocols. For example, the protocol processing circuitry may operate LTE protocol entities or 5G NR protocol entities, or both, when the baseband circuitry **610** or RF circuitry **606**, or both, are part of mm Wave communication circuitry or some other suitable cellular communication circuitry. In this example, the protocol processing circuitry can operate MAC, RLC, PDCP, SDAP, RRC, and NAS functions. In some implementations, the protocol processing circuitry may operate one or more IEEE-based protocols when the baseband circuitry **610** or RF circuitry **606**, or both, are part of a Wi-Fi communication system. In this example, the protocol processing circuitry can operate Wi-Fi MAC and logical link control (LLC) functions. The protocol processing circuitry may include one or more memory structures (e.g., **604G**) to store program code and data for operating the protocol functions, as well as one or more processing cores to execute the program code and perform various operations using the data. The baseband circuitry **610** may also support radio communications for more than one wireless protocol.

The various hardware elements of the baseband circuitry **610** discussed herein may be implemented, for example, as a solder-down substrate including one or more integrated circuits (ICs), a single packaged IC soldered to a main circuit board or a multi-chip module containing two or more ICs. In some implementations, the components of the baseband circuitry **610** may be suitably combined in a single chip or chipset, or disposed on a same circuit board. In some implementations, some or all of the constituent components of the baseband circuitry **610** and RF circuitry **606** may be implemented together such as, for example, a system on a chip (SoC) or System-in-Package (SiP). In some implementations, some or all of the constituent components of the baseband circuitry **610** may be implemented as a separate SoC that is communicatively coupled with and RF circuitry **606** (or multiple instances of RF circuitry **606**). In some implementations, some or all of the constituent components of the baseband circuitry **610** and the application circuitry **405**, **505** may be implemented together as individual SoCs mounted to a same circuit board (e.g., a “multi-chip package”).

In some implementations, the baseband circuitry **610** may provide for communication compatible with one or more radio technologies. The RF circuitry **606** may enable communication with wireless networks using modulated electromagnetic radiation through a non-solid medium. In some implementations, the RF circuitry **606** may include switches, filters, or amplifiers, among other components, to facilitate the communication with the wireless network. The RF circuitry **606** may include a receive signal path, which may include circuitry to down-convert RF signals received from the FEM circuitry **608** and provide baseband signals to the baseband circuitry **610**. The RF circuitry **606** may also include a transmit signal path, which may include circuitry to up-convert baseband signals provided by the baseband

circuitry **610** and provide RF output signals to the FEM circuitry **608** for transmission.

The receive signal path of the RF circuitry **606** includes mixer circuitry **606a**, amplifier circuitry **606b** and filter circuitry **606c**. In some implementations, the transmit signal path of the RF circuitry **606** may include filter circuitry **606c** and mixer circuitry **606a**. The RF circuitry **606** also includes synthesizer circuitry **606d** for synthesizing a frequency for use by the mixer circuitry **606a** of the receive signal path and the transmit signal path. In some implementations, the mixer circuitry **606a** of the receive signal path may be configured to down-convert RF signals received from the FEM circuitry **608** based on the synthesized frequency provided by synthesizer circuitry **606d**. The amplifier circuitry **606b** may be configured to amplify the down-converted signals and the filter circuitry **606c** may be a low-pass filter (LPF) or band-pass filter (BPF) configured to remove unwanted signals from the down-converted signals to generate output baseband signals. Output baseband signals may be provided to the baseband circuitry **610** for further processing. In some implementations, the output baseband signals may be zero-frequency baseband signals, although this is not a requirement. In some implementations, the mixer circuitry **606a** of the receive signal path may comprise passive mixers.

In some implementations, the mixer circuitry **606a** of the transmit signal path may be configured to up-convert input baseband signals based on the synthesized frequency provided by the synthesizer circuitry **606d** to generate RF output signals for the FEM circuitry **608**. The baseband signals may be provided by the baseband circuitry **610** and may be filtered by filter circuitry **606c**.

In some implementations, the mixer circuitry **606a** of the receive signal path and the mixer circuitry **606a** of the transmit signal path may include two or more mixers and may be arranged for quadrature downconversion and upconversion, respectively. In some implementations, the mixer circuitry **606a** of the receive signal path and the mixer circuitry **606a** of the transmit signal path may include two or more mixers and may be arranged for image rejection (e.g., Hartley image rejection). In some implementations, the mixer circuitry **606a** of the receive signal path and the mixer circuitry **606a** of the transmit signal path may be arranged for direct downconversion and direct upconversion, respectively. In some implementations, the mixer circuitry **606a** of the receive signal path and the mixer circuitry **606a** of the transmit signal path may be configured for super-heterodyne operation.

In some implementations, the output baseband signals and the input baseband signals may be analog baseband signals. In some implementations, the output baseband signals and the input baseband signals may be digital baseband signals, and the RF circuitry **606** may include analog-to-digital converter (ADC) and digital-to-analog converter (DAC) circuitry and the baseband circuitry **610** may include a digital baseband interface to communicate with the RF circuitry **606**. In some dual-mode examples, a separate radio IC circuitry may be provided for processing signals for each spectrum, although the techniques described here are not limited in this respect.

In some implementations, the synthesizer circuitry **606d** may be a fractional-N synthesizer or a fractional N/N+1 synthesizer, although other types of frequency synthesizers may be used. For example, synthesizer circuitry **606d** may be a delta-sigma synthesizer, a frequency multiplier, or a synthesizer comprising a phase-locked loop with a frequency divider. The synthesizer circuitry **606d** may be configured to synthesize an output frequency for use by the

mixer circuitry **606a** of the RF circuitry **606** based on a frequency input and a divider control input. In some implementations, the synthesizer circuitry **606d** may be a fractional $N/N+1$ synthesizer.

In some implementations, frequency input may be provided by a voltage controlled oscillator (VCO), although that is not a requirement. Divider control input may be provided by either the baseband circuitry **610** or the application circuitry **405/505** depending on the desired output frequency. In some implementations, a divider control input (e.g., N) may be determined from a look-up table based on a channel indicated by the application circuitry **405**, **505**.

The synthesizer circuitry **606d** of the RF circuitry **606** may include a divider, a delay-locked loop (DLL), a multiplexer and a phase accumulator. In some implementations, the divider may be a dual modulus divider (DMD) and the phase accumulator may be a digital phase accumulator (DPA). In some implementations, the DMD may be configured to divide the input signal by either N or $N+1$ (e.g., based on a carry out) to provide a fractional division ratio. In some implementations, the DLL may include a set of cascaded, tunable, delay elements, a phase detector, a charge pump and a D-type flip-flop. The delay elements may be configured to break a VCO period up into N_d equal packets of phase, where N_d is the number of delay elements in the delay line. In this way, the DLL provides negative feedback to help ensure that the total delay through the delay line is one VCO cycle.

In some implementations, synthesizer circuitry **606d** may be configured to generate a carrier frequency as the output frequency, while in other examples, the output frequency may be a multiple of the carrier frequency (e.g., twice the carrier frequency, four times the carrier frequency) and used in conjunction with quadrature generator and divider circuitry to generate multiple signals at the carrier frequency with multiple different phases with respect to each other. In some implementations, the output frequency may be a LO frequency (fLO). In some implementations, the RF circuitry **606** may include an IQ or polar converter.

The FEM circuitry **608** may include a receive signal path, which may include circuitry configured to operate on RF signals received from antenna array **611**, amplify the received signals and provide the amplified versions of the received signals to the RF circuitry **606** for further processing. The FEM circuitry **608** may also include a transmit signal path, which may include circuitry configured to amplify signals for transmission provided by the RF circuitry **606** for transmission by one or more of antenna elements of antenna array **611**. The amplification through the transmit or receive signal paths may be done solely in the RF circuitry **606**, solely in the FEM circuitry **608**, or in both the RF circuitry **606** and the FEM circuitry **608**.

In some implementations, the FEM circuitry **608** may include a TX/RX switch to switch between transmit mode and receive mode operation. The FEM circuitry **608** may include a receive signal path and a transmit signal path. The receive signal path of the FEM circuitry **608** may include an LNA to amplify received RF signals and provide the amplified received RF signals as an output (e.g., to the RF circuitry **606**). The transmit signal path of the FEM circuitry **608** may include a power amplifier (PA) to amplify input RF signals (e.g., provided by RF circuitry **606**), and one or more filters to generate RF signals for subsequent transmission by one or more antenna elements of the antenna array **611**.

The antenna array **611** comprises one or more antenna elements, each of which is configured convert electrical signals into radio waves to travel through the air and to

convert received radio waves into electrical signals. For example, digital baseband signals provided by the baseband circuitry **610** is converted into analog RF signals (e.g., modulated waveform) that will be amplified and transmitted using the antenna elements of the antenna array **611** including one or more antenna elements (not shown). The antenna elements may be omnidirectional, directional, or a combination thereof. The antenna elements may be formed in a multitude of arrangements as are known and/or discussed herein. The antenna array **611** may comprise microstrip antennas or printed antennas that are fabricated on the surface of one or more printed circuit boards. The antenna array **611** may be formed as a patch of metal foil (e.g., a patch antenna) in a variety of shapes, and may be coupled with the RF circuitry **606** and/or FEM circuitry **608** using metal transmission lines or the like.

Processors of the application circuitry **405/505** and processors of the baseband circuitry **610** may be used to execute elements of one or more instances of a protocol stack. For example, processors of the baseband circuitry **610**, alone or in combination, may execute Layer 3, Layer 2, or Layer 1 functionality, while processors of the application circuitry **405**, **505** may utilize data (e.g., packet data) received from these layers and further execute Layer 4 functionality (e.g., TCP and UDP layers). As referred to herein, Layer 3 may comprise a RRC layer, described in further detail below. As referred to herein, Layer 2 may comprise a MAC layer, an RLC layer, and a PDCP layer, described in further detail below. As referred to herein, Layer 1 may comprise a PHY layer of a UE/RAN node, described in further detail below.

FIG. 7 illustrates example components of communication circuitry **700**. In some implementations, the communication circuitry **700** may be implemented as part of the system **400** or the platform **500** shown in FIGS. 4 and 5. The communication circuitry **700** may be communicatively coupled (e.g., directly or indirectly) to one or more antennas, such as antennas **711a**, **711b**, **711c**, and **711d**. In some implementations, the communication circuitry **700** includes or is communicatively coupled to dedicated receive chains, processors, or radios, or combinations of them, for multiple RATs (e.g., a first receive chain for LTE and a second receive chain for 5G NR). For example, as shown in FIG. 7, the communication circuitry **700** includes a modem **710** and a modem **720**, which may correspond to or be a part of the baseband circuitry **410** and **510** of FIGS. 4 and 5. The modem **710** may be configured for communications according to a first RAT, such as LTE or LTE-A, and the modem **720** may be configured for communications according to a second RAT, such as 5G NR. In some implementations, a processor **705**, such as an application processor can interface with the modems **710**, **720**.

The modem **710** includes one or more processors **712** and a memory **716** in communication with the processors **712**. The modem **710** is in communication with a radio frequency (RF) front end **730**, which may correspond to or be a part of to the RFEM **415** and **515** of FIGS. 4 and 5. The RF front end **730** may include circuitry for transmitting and receiving radio signals. For example, the RF front end **730** includes RX circuitry **732** and TX circuitry **734**. In some implementations, the receive circuitry **732** is in communication with a DL front end **752**, which may include circuitry for receiving radio signals from one or more antennas **711a**. The transmit circuitry **734** is in communication with a UL front end **754**, which is coupled with one or more antennas **711b**.

Similarly, the modem **720** includes one or more processors **722** and a memory **726** in communication with the one or more processors **722**. The modem **720** is in communica-

tion with an RF front end **740**, which may correspond to or be a part of to the RFEM **415** and **515** of FIGS. **4** and **5**. The RF front end **740** may include circuitry for transmitting and receiving radio signals. For example, the RF front end **740** includes receive circuitry **742** and transmit circuitry **744**. In some implementations, the receive circuitry **742** is in communication with a DL front end **760**, which may include circuitry for receiving radio signals from one or more antennas **711c**. The transmit circuitry **744** is in communication with a UL front end **765**, which is coupled with one or more antennas **711d**. In some implementations, one or more front-ends can be combined. For example, a RF switch can selectively couple the modems **710**, **720** to a single UL front end **772** for transmitting radio signals using one or more antennas.

The modem **710** may include hardware and software components for time division multiplexing UL data (e.g., for NSA NR operations), as well as the various other techniques described herein. The processors **712** may include one or more processing elements configured to implement various features described herein, such as by executing program instructions stored on the memory **716** (e.g., a non-transitory computer-readable memory medium). In some implementations, the processor **712** may be configured as a programmable hardware element, such as a FPGA or an ASIC. In some implementations, the processors **712** may include one or more ICs that are configured to perform the functions of processors **712**. For example, each IC may include circuitry configured to perform the functions of processors **712**.

The modem **720** may include hardware and software components for time division multiplexing UL data (e.g., for NSA NR operations), as well as the various other techniques described herein. The processors **722** may include one or more processing elements configured to implement various features described herein, such as by executing instructions stored on the memory **726** (e.g., a non-transitory computer-readable memory medium). In some implementations, the processor **722** may be configured as a programmable hardware element, such as a FPGA or an ASIC. In some implementations, the processor **722** may include one or more ICs that are configured to perform the functions of processors **722**.

FIG. **8** illustrates various protocol functions that may be implemented in a wireless communication device. In particular, FIG. **8** includes an arrangement **800** showing interconnections between various protocol layers/entities. The following description of FIG. **8** is provided for various protocol layers and entities that operate in conjunction with the 5G NR system standards and the LTE system standards, but some or all of the aspects of FIG. **8** may be applicable to other wireless communication network systems as well.

The protocol layers of arrangement **800** may include one or more of PHY **810**, MAC **820**, RLC **830**, PDCP **840**, SDAP **847**, RRC **855**, and NAS layer **857**, in addition to other higher layer functions not illustrated. The protocol layers may include one or more service access points (e.g., items **859**, **856**, **850**, **849**, **845**, **835**, **825**, and **815** in FIG. **8**) that may provide communication between two or more protocol layers.

The PHY **810** may transmit and receive physical layer signals **805** that may be received from or transmitted to one or more other communication devices. The physical layer signals **805** may include one or more physical channels, such as those discussed herein. The PHY **810** may further perform link adaptation or adaptive modulation and coding (AMC), power control, cell search (e.g., for initial synchronization and handover purposes), and other measurements used by

higher layers, such as the RRC **855**. The PHY **810** may still further perform error detection on the transport channels, forward error correction (FEC) coding and decoding of the transport channels, modulation and demodulation of physical channels, interleaving, rate matching, mapping onto physical channels, and MIMO antenna processing. In some implementations, an instance of PHY **810** may process requests from and provide indications to an instance of MAC **820** using one or more PHY-SAP **815**. In some implementations, requests and indications communicated using PHY-SAP **815** may comprise one or more transport channels.

Instance(s) of MAC **820** may process requests from, and provide indications to, an instance of RLC **830** using one or more MAC-SAPs **825**. These requests and indications communicated using the MAC-SAP **825** may include one or more logical channels. The MAC **820** may perform mapping between the logical channels and transport channels, multiplexing of MAC SDUs from one or more logical channels onto transport blocks (TBs) to be delivered to PHY **810** using the transport channels, de-multiplexing MAC SDUs to one or more logical channels from TBs delivered from the PHY **810** using transport channels, multiplexing MAC SDUs onto TBs, scheduling information reporting, error correction through HARQ, and logical channel prioritization.

Instance(s) of RLC **830** may process requests from and provide indications to an instance of PDCP **840** using one or more radio link control service access points (RLC-SAP) **835**. These requests and indications communicated using RLC-SAP **835** may include one or more RLC channels. The RLC **830** may operate in a plurality of modes of operation, including: Transparent Mode (TM), Unacknowledged Mode (UM), and Acknowledged Mode (AM). The RLC **830** may execute transfer of upper layer protocol data units (PDUs), error correction through automatic repeat request (ARQ) for AM data transfers, and concatenation, segmentation and reassembly of RLC SDUs for UM and AM data transfers. The RLC **830** may also execute re-segmentation of RLC data PDUs for AM data transfers, reorder RLC data PDUs for UM and AM data transfers, detect duplicate data for UM and AM data transfers, discard RLC SDUs for UM and AM data transfers, detect protocol errors for AM data transfers, and perform RLC re-establishment.

Instance(s) of PDCP **840** may process requests from and provide indications to instance(s) of RRC **855** or instance(s) of SDAP **847**, or both, using one or more packet data convergence protocol service access points (PDCP-SAP) **845**. These requests and indications communicated using PDCP-SAP **845** may include one or more radio bearers. The PDCP **840** may execute header compression and decompression of IP data, maintain PDCP Sequence Numbers (SNs), perform in-sequence delivery of upper layer PDUs at re-establishment of lower layers, eliminate duplicates of lower layer SDUs at re-establishment of lower layers for radio bearers mapped on RLC AM, cipher and decipher control plane data, perform integrity protection and integrity verification of control plane data, control timer-based discard of data, and perform security operations (e.g., ciphering, deciphering, integrity protection, or integrity verification).

Instance(s) of SDAP **847** may process requests from and provide indications to one or more higher layer protocol entities using one or more SDAP-SAP **849**. These requests and indications communicated using SDAP-SAP **849** may include one or more QoS flows. The SDAP **847** may map QoS flows to data radio bearers (DRBs), and vice versa, and

may also mark QoS flow identifiers (QFIs) in DL and UL packets. A single SDAP entity **847** may be configured for an individual PDU session. In the UL direction, the NG-RAN **110** may control the mapping of QoS Flows to DRB(s) in two different ways, reflective mapping or explicit mapping. For reflective mapping, the SDAP **847** of a UE **101** may monitor the QFIs of the DL packets for each DRB, and may apply the same mapping for packets flowing in the UL direction. For a DRB, the SDAP **847** of the UE **101** may map the UL packets belonging to the QoS flows(s) corresponding to the QoS flow ID(s) and PDU session observed in the DL packets for that DRB. To enable reflective mapping, the NG-RAN **310** may mark DL packets over the Uu interface with a QoS flow ID. The explicit mapping may involve the RRC **855** configuring the SDAP **847** with an explicit QoS flow to DRB mapping rule, which may be stored and followed by the SDAP **847**. In some implementations, the SDAP **847** may only be used in NR implementations and may not be used in LTE implementations.

The RRC **855** may configure, using one or more management service access points (M-SAP), aspects of one or more protocol layers, which may include one or more instances of PHY **810**, MAC **820**, RLC **830**, PDCP **840** and SDAP **847**. In some implementations, an instance of RRC **855** may process requests from and provide indications to one or more NAS entities **857** using one or more RRC-SAPs **856**. The main services and functions of the RRC **855** may include broadcast of system information (e.g., included in master information blocks (MIBs) or system information blocks (SIBs) related to the NAS), broadcast of system information related to the access stratum (AS), paging, establishment, maintenance and release of an RRC connection between the UE **101** and RAN **110** (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), establishment, configuration, maintenance and release of point to point Radio Bearers, security functions including key management, inter-RAT mobility, and measurement configuration for UE measurement reporting. The MIBs and SIBs may comprise one or more information elements (IEs), which may each comprise individual data fields or data structures.

The NAS **857** may form the highest stratum of the control plane between the UE **101** and the AMF **321**. The NAS **857** may support the mobility of the UEs **101** and the session management procedures to establish and maintain IP connectivity between the UE **101** and a P-GW in LTE systems. In some implementations, one or more protocol entities of arrangement **800** may be implemented in UEs **101**, RAN nodes **111**, AMF **321** in NR implementations or MME **221** in LTE implementations, UPF **302** in NR implementations or S-GW **222** and P-GW **223** in LTE implementations, or the like to be used for control plane or user plane communications protocol stack between the aforementioned devices. In some implementations, one or more protocol entities that may be implemented in one or more of UE **101**, gNB **111**, AMF **321**, among others, may communicate with a respective peer protocol entity that may be implemented in or on another device using the services of respective lower layer protocol entities to perform such communication. In some implementations, a gNB-CU of the gNB **111** may host the RRC **855**, SDAP **847**, and PDCP **840** of the gNB that controls the operation of one or more gNB-DUs, and the gNB-DUs of the gNB **111** may each host the RLC **830**, MAC **820**, and PHY **810** of the gNB **111**.

In some implementations, a control plane protocol stack may include, in order from highest layer to lowest layer,

NAS **857**, RRC **855**, PDCP **840**, RLC **830**, MAC **820**, and PHY **810**. In this example, upper layers **860** may be built on top of the NAS **857**, which includes an IP layer **861**, an SCTP **862**, and an application layer signaling protocol (AP) **863**.

In some implementations, such as NR implementations, the AP **863** may be an NG application protocol layer (NGAP or NG-AP) **863** for the NG interface **113** defined between the NG-RAN node **111** and the AMF **321**, or the AP **863** may be an Xn application protocol layer (XnAP or Xn-AP) **863** for the Xn interface **112** that is defined between two or more RAN nodes **111**. The NG-AP **863** may support the functions of the NG interface **113** and may comprise elementary procedures (EPS). An NG-AP EP may be a unit of interaction between the NG-RAN node **111** and the AMF **321**. The NG-AP **863** services may include two groups: UE-associated services (e.g., services related to a UE **101**) and non-UE-associated services (e.g., services related to the whole NG interface instance between the NG-RAN node **111** and AMF **321**). These services may include functions such as, but not limited to: a paging function for the sending of paging requests to NG-RAN nodes **111** involved in a particular paging area; a UE context management function for allowing the AMF **321** to establish, modify, or release a UE context in the AMF **321** and the NG-RAN node **111**; a mobility function for UEs **101** in ECM-CONNECTED mode for intra-system HOs to support mobility within NG-RAN and inter-system HOs to support mobility from/to EPS systems; a NAS Signaling Transport function for transporting or rerouting NAS messages between UE **101** and AMF **321**; a NAS node selection function for determining an association between the AMF **321** and the UE **101**; NG interface management function(s) for setting up the NG interface and monitoring for errors over the NG interface; a warning message transmission function for providing means to transfer warning messages using NG interface or cancel ongoing broadcast of warning messages; a configuration transfer function for requesting and transferring of RAN configuration information (e.g., SON information or performance measurement (PM) data) between two RAN nodes **111** using CN **120**, or combinations of them, among others.

The XnAP **863** may support the functions of the Xn interface **112** and may comprise XnAP basic mobility procedures and XnAP global procedures. The XnAP basic mobility procedures may comprise procedures used to handle UE mobility within the NG RAN **111** (or E-UTRAN **210**), such as handover preparation and cancellation procedures, SN Status Transfer procedures, UE context retrieval and UE context release procedures, RAN paging procedures, or dual connectivity related procedures, among others. The XnAP global procedures may comprise procedures that are not related to a specific UE **101**, such as Xn interface setup and reset procedures, NG-RAN update procedures, or cell activation procedures, among others.

In LTE implementations, the AP **863** may be an S1 Application Protocol layer (S1-AP) **863** for the S1 interface **113** defined between an E-UTRAN node **111** and an MME, or the AP **863** may be an X2 application protocol layer (X2AP or X2-AP) **863** for the X2 interface **112** that is defined between two or more E-UTRAN nodes **111**. The S1 Application Protocol layer (S1-AP) **863** may support the functions of the S1 interface, and similar to the NG-AP discussed previously, the S1-AP may include S1-APEPs. An S1-AP EP may be a unit of interaction between the E-UTRAN node **111** and an MME **221** within a LTE CN **120**. The S1-AP **863** services may comprise two groups: UE-associated services and non UE-associated services.

These services perform functions including, but not limited to: E-UTRAN Radio Access Bearer (E-RAB) management, UE capability indication, mobility, NAS signaling transport, RAN Information Management (RIM), and configuration transfer.

The X2AP **863** may support the functions of the X2 interface **112** and may include X2AP basic mobility procedures and X2AP global procedures. The X2AP basic mobility procedures may include procedures used to handle UE mobility within the E-UTRAN **120**, such as handover preparation and cancellation procedures, SN Status Transfer procedures, UE context retrieval and UE context release procedures, RAN paging procedures, or dual connectivity related procedures, among others. The X2AP global procedures may comprise procedures that are not related to a specific UE **101**, such as X2 interface setup and reset procedures, load indication procedures, error indication procedures, or cell activation procedures, among others.

The SCTP layer (alternatively referred to as the SCTP/IP layer) **862** may provide guaranteed delivery of application layer messages (e.g., NGAP or XnAP messages in NR implementations, or S1-AP or X2AP messages in LTE implementations). The SCTP **862** may ensure reliable delivery of signaling messages between the RAN node **111** and the AMF **321/MME 221** based in part on the IP protocol, supported by the IP **861**. The Internet Protocol layer (IP) **861** may be used to perform packet addressing and routing functionality. In some implementations the IP layer **861** may use point-to-point transmission to deliver and convey PDUs. In this regard, the RAN node **111** may include L2 and L1 layer communication links (e.g., wired or wireless) with the MME/AMF to exchange information.

In some implementations, a user plane protocol stack may include, in order from highest layer to lowest layer, SDAP **847**, PDCP **840**, RLC **830**, MAC **820**, and PHY **810**. The user plane protocol stack may be used for communication between the UE **101**, the RAN node **111**, and UPF **302** in NR implementations or an S-GW **222** and P-GW **223** in LTE implementations. In this example, upper layers **851** may be built on top of the SDAP **847**, and may include a user datagram protocol (UDP) and IP security layer (UDP/IP) **852**, a General Packet Radio Service (GPRS) Tunneling Protocol for the user plane layer (GTP-U) **853**, and a User Plane PDU layer (UP PDU) **863**.

The transport network layer **854** (also referred to as a “transport layer”) may be built on IP transport, and the GTP-U **853** may be used on top of the UDP/IP layer **852** (comprising a UDP layer and IP layer) to carry user plane PDUs (UP-PDUs). The IP layer (also referred to as the “Internet layer”) may be used to perform packet addressing and routing functionality. The IP layer may assign IP addresses to user data packets in any of IPv4, IPv6, or PPP formats, for example.

The GTP-U **853** may be used for carrying user data within the GPRS core network and between the radio access network and the core network. The user data transported can be packets in any of IPv4, IPv6, or PPP formats, for example. The UDP/IP **852** may provide checksums for data integrity, port numbers for addressing different functions at the source and destination, and encryption and authentication on the selected data flows. The RAN node **111** and the S-GW **222** may utilize an S1-U interface to exchange user plane data using a protocol stack comprising an L1 layer (e.g., PHY **810**), an L2 layer (e.g., MAC **820**, RLC **830**, PDCP **840**, and/or SDAP **847**), the UDP/IP layer **852**, and the GTP-U **853**. The S-GW **222** and the P-GW **223** may utilize an S5/S8a interface to exchange user plane data using

a protocol stack comprising an L1 layer, an L2 layer, the UDP/IP layer **852**, and the GTP-U **853**. As discussed previously, NAS protocols may support the mobility of the UE **101** and the session management procedures to establish and maintain IP connectivity between the UE **101** and the P-GW **223**.

Moreover, although not shown by FIG. **8**, an application layer may be present above the AP **863** and/or the transport network layer **854**. The application layer may be a layer in which a user of the UE **101**, RAN node **111**, or other network element interacts with software applications being executed, for example, by application circuitry **405** or application circuitry **505**, respectively. The application layer may also provide one or more interfaces for software applications to interact with communications systems of the UE **101** or RAN node **111**, such as the baseband circuitry **610**. In some implementations, the IP layer or the application layer, or both, may provide the same or similar functionality as layers 5-7, or portions thereof, of the Open Systems Interconnection (OSI) model (e.g., OSI Layer 7—the application layer, OSI Layer 6—the presentation layer, and OSI Layer 5—the session layer).

NFV architectures and infrastructures may be used to virtualize one or more NFs, alternatively performed by proprietary hardware, onto physical resources comprising a combination of industry-standard server hardware, storage hardware, or switches. In other words, NFV systems can be used to execute virtual or reconfigurable implementations of one or more EPC components and functions.

FIG. **9** illustrates a block diagram of example of a computer system that includes components for reading instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and performing any one or more of the techniques described herein. In this example, FIG. **9** shows a diagrammatic representation of hardware resources **900** including one or more processors (or processor cores) **910**, one or more memory or storage devices **920**, and one or more communication resources **930**, each of which may be communicatively coupled using a bus **940**. For implementations where node virtualization (e.g., NFV) is utilized, a hypervisor **902** may be executed to provide an execution environment for one or more network slices or sub-slices to utilize the hardware resources **900**.

The processors **910** may include a processor **912** and a processor **914**. The processor(s) **910** may be, for example, a CPU, a RISC processor, a CISC processor, a graphics processing unit (GPU), a DSP such as a baseband processor, an ASIC, an FPGA, a radio-frequency integrated circuit (RFIC), another processor (including those discussed herein), or any suitable combination thereof. The memory/storage devices **920** may include main memory, disk storage, or any suitable combination thereof. The memory/storage devices **920** may include, but are not limited to, any type of volatile or nonvolatile memory such as DRAM, SRAM, EPROM, EEPROM, Flash memory, or solid-state storage, or combinations of them, among others.

The communication resources **930** may include interconnection or network interface components or other suitable devices to communicate with one or more peripheral devices **904** or one or more databases **906** using a network **908**. For example, the communication resources **930** may include wired communication components (e.g., for coupling using USB), cellular communication components, NFC components, Bluetooth® (or Bluetooth® Low Energy) components, Wi-Fi components, and other communication components.

Instructions 950 can include software, a program, an application, an applet, an app, or other executable code for causing at least any of the processors 910 to perform any one or more of the methodologies discussed herein. The instructions 950 may reside, completely or partially, within at least one of the processors 910 (e.g., within the processor's cache memory), the memory/storage devices 920, or any suitable combination thereof. Furthermore, any portion of the instructions 950 may be transferred to the hardware resources 900 from any combination of the peripheral devices 904 or the databases 906. Accordingly, the memory of processors 910, the memory/storage devices 920, the peripheral devices 904, and the databases 906 are examples of computer-readable and machine-readable media.

FIG. 10 illustrates a diagram of an example of a wireless communication system 1001 that includes a RAN node 1005 and UE 1020. The wireless communication system 1001 can use MIMO techniques to transmit data between user devices and base stations. MIMO techniques including using multiple antennas at the base station, user device, or both. MIMO techniques can include beamforming techniques, single user MIMO (SU-MIMO), multi-user MIMO (MU-MIMO), and others. MIMO can provide increased communication bandwidth. The system 1001 can be based on NR. A RAN node 1005 can transmit or receive control and data channels to or from a UE 1020. The RAN node 1005 can be equipped with antennas such as two or more sub-arrays or panels for MIMO communications. In some implementations, the UE 1020 can be equipped with two or more sub-arrays or panels. In some implementations, the UE 1020 can transmit or receive control and data channels using two or more panels simultaneously to improve the link budget. Further, transmitting devices including the RAN node 1005 and the UE 1020 can transmit reference signals such as DMRS in addition to a data channel to help a receiving device decode a data channel such as PDSCH or PUSCH. In some implementations, two or more antennas of a transmitting device can transmit respective reference signals to help a receiving device decode a data channel that is transmitted on the two or more antennas.

In addition to transmitting user data, base stations can provide control signaling to user devices. Various types of control signals include scheduling information and decoding information such as DCI messages. A user device can use the scheduling information and decoding information to receive and decode user data transmitted on a downlink channel such as PDSCH. The user's data can be multiplexed in time, space, frequency, or a combination thereof. PDSCH transmissions, for example, can be scheduled by one or more DCI messages, which can also schedule uplink transmissions such as PUSCH transmissions that provide HARQ feedback for the PDSCH transmissions. A DCI message can include among other things one or more time and frequency resource assignments for receiving PDSCH transmissions, DMRS antenna port information, a modulation and coding scheme indicator, HARQ process number, and control information for a corresponding PUSCH transmission such as power control, PUSCH resource indicator (PRI), and a PDSCH-to-HARQ_feedback timing indicator.

This disclosure provides, among other things, systems and techniques for NR systems to indicate DMRS antenna ports to UEs for Ultra Reliable Low-Latency Communications (URLLC) transmissions in DL and UL using different bit-widths for the DCI indication field. Examples of corresponding entries of DMRS antenna port indication tables are also provided. This disclosure also provides a mapping of DMRS antenna port indication values from 3GPP NR

antenna port tables to new values for NR URLLC transmission/reception based on a configured bit-width of the antenna port indication field in DCI. One or more of the disclosed techniques can reduce DCI overhead and increase system efficiency.

In Rel-15 NR, the DMRS antenna port indication field in DCI formats 0_1 and 1_1 for uplink and downlink transmissions respectively are of fixed bit-width which is determined by the RRC parameters dmrs-Type and maxLength from the PDSCH and PUSCH configurations as well as the transmit rank indicator (TRI) field for UL. For NR URLLC transmission, the bit-width of the antenna port indication fields in the DCI can be further configurable. In some implementations, RRC signaling is used to configure the number of bits for the bit-field for DCI formats 0_1 and 1_1 for UL and DL respectively or for a new DCI format. In some implementations, RRC signaling is used to configure the number of antenna ports (APs) to be assumed for the above two choices of DCI formats for NR URLLC transmission and/or reception.

A gNB can provide a downlink DMRS antenna port indication. In some implementations, for a single codeword transmission for the case when a non-fallback DCI is used and dmrs-Type=1; and the maximum number of front-loaded DMRS symbols is limited to a single symbol, e.g., RRC configured value of maxLength=1, and the configured bit-width of the antenna port indication field in DCI is 4 bits, the antenna port indication value can be parsed in accordance with Table 1 to determine a DMRS antenna port configuration. The non-fallback DCI can be based on a DCI format 1_1 or a new DCI format.

TABLE 1

Antenna port(s) (1000 + DMRS port), dmrs-Type = 1, maxLength = 1 One Codeword: Codeword 0 enabled, Codeword 1 disabled		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0
1	1	1
2	1	0, 1
3	2	0
4	2	1
5	2	2
6	2	3
7	2	0, 1
8	2	2, 3
9	2	0-2
10	2	0-3
11	2	0, 2
12-15	Reserved	Reserved

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 0-bit, the UE is expected to assume the default value of the antenna port indication is value 3 from Table 1. In some implementations, the UE may assume that the default value for 0-bit indication is value 0 from Table 1.

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 1-bit, the UE is expected to assume that the bit value 0 corresponds to value 3 in Table 1 and bit value 1 corresponds to value 7 in Table 1, where a SU-MIMO transmission may be assumed. In some implementations, bit value 0 may corresponds to value 3 in Table 1, whereas bit value 1 corresponds to value 4 in Table 1.

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In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 2 bits, the UE is expected to assume the mapping provided in Table 2 of the 2-bit DCI indication to values from Table 1.

TABLE 2

DCI to Antenna Port Indication Mapping for 2 bit indication for dmrs-Type = 1, maxLength = 1			
DCI Bits	Value (From Table 1)	Number of DMRS CDM group(s) without data	DMRS port(s)
00	3	2	0
01	4	2	1
10	5	2	2
11	6	2	3

In some implementations, the UE may assume the mapping provided in Table 3 of the 2 bit DCI indication to values from Table 1. This option can allow for signaling of co-scheduled DMRS ports, which can be useful for MU-MIMO-based scheduling.

TABLE 3

DCI to Antenna Port Indication Mapping for 2 bit indication for dmrs-Type = 1, maxLength = 1			
DCI Bits	Value (From Table 1)	Number of DMRS CDM group(s) without data	DMRS port(s)
00	3	2	0
01	4	2	1
10	7	2	0, 1
11	8	2	2, 3

In some implementations, for a single codeword transmission for the case when a non-fallback DCI is used and dmrs-Type=2; and the maximum number of front loaded DMRS symbols is limited to a single symbol, e.g., RRC configured value of maxLength=1, and the configured bit-width of the antenna port indication field in DCI is 5 bits, the antenna port indication value can be parsed in accordance with Table 1 to determine a DMRS antenna port configuration. The non-fallback DCI can be based on a DCI format 1_1 or a new DCI format.

TABLE 4

Antenna port(s) (1000 + DMRS port), dmrs-Type = 2, maxLength = 1					
One codeword: Codeword 0 enabled, Codeword 1 disabled			Two codewords: Codeword 0 enabled, Codeword 1 enabled		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0	0	3	0-4
1	1	0, 1	1	3	0-5
2	1	0, 1	2-31	reserved	reserved
3	2	0			
4	2	1			
5	2	2			
6	2	3			

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TABLE 4-continued

Antenna port(s) (1000 + DMRS port), dmrs-Type = 2, maxLength = 1					
One codeword: Codeword 0 enabled, Codeword 1 disabled			Two codewords: Codeword 0 enabled, Codeword 1 enabled		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Value	Number of DMRS CDM group(s) without data	DMRS port(s)
7	2	0, 1			
8	2	2, 3			
9	2	0-2			
10	2	0-3			
11	3	0			
12	3	1			
13	3	2			
14	3	3			
15	3	4			
16	3	5			
17	3	0, 1			
18	3	2, 3			
19	3	4, 5			
20	3	0-2			
21	3	3-5			
22	3	0-3			
23	2	0, 2			
24-31	Reserved	Reserved			

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 0-bit, the UE is expected to assume the default value of the antenna port indication is value 11 from Table 4. In some implementations, the UE may assume that the default value for 0-bit indication is value 0 from Table 4. In some implementations, the UE may assume that the default value for 0-bit indication is value 3 from Table 4.

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 1-bit, the UE is expected to assume that bit 0 corresponds to value 11 and bit 1 corresponds to value 17 from Table 4. In some implementations, the UE may assume that bit 0 corresponds to value 11 and bit 1 corresponds to value 12 from Table 4.

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 2-bits, the UE is expected to assume the mapping provided in Table of the 2 bit DCI indication to values from Table 4.

TABLE 5

DCI to Antenna Port Indication Mapping for 2 bit indication for dmrs-Type = 2, maxLength = 1			
DCI Bits	Value (From Table 4)	Number of DMRS CDM group(s) without data	DMRS port(s)
00	11	3	0
01	12	3	1
10	13	3	2
11	14	3	3

In some implementations, the UE may also assume the mapping provided in Table 6 of the 2 bit DCI indication to values from Table 4.

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TABLE 6

DCI to Antenna Port Indication Mapping for 2 bit indication for dmrs-Type = 2, maxLength = 1			
DCI Bits	Value (From Table 4)	Number of DMRS CDM group(s) without data	DMRS port(s)
00	11	3	0
01	12	3	1
10	17	3	0, 1
11	18	3	2, 3

In some implementations, the UE may also assume the mapping provided in Table 7 of the 2 bit DCI indications to values from Table 4.

TABLE 7

DCI to Antenna Port Indication Mapping for 2 bit indication for dmrs-Type = 2, maxLength = 1			
DCI Bits	Value (From Table 4)	Number of DMRS CDM group(s) without data	DMRS port(s)
00	11	3	0
01	17	3	0, 1
10	20	3	0-2
11	22	3	0-3

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 4-bits, the UE may also assume the mapping provided in Table 8 of the 4-bit DCI antenna port indication values to values from Table 4.

TABLE 8

DCI to Antenna Port Indication Mapping for 4-bit indication for dmrs-Type = 2, maxLength = 1			
DCI Value	Value (From Table 4)	Number of DMRS CDM group(s) without data	DMRS port(s)
0	3	2	0
1	4	2	1
2	5	2	2
3	6	2	3
4	7	2	0, 1
5	8	2	2, 3
6	11	3	0
7	12	3	1
8	13	3	2
9	14	3	3
10	15	3	4
11	16	3	5
12	17	3	0, 1
13	18	3	2, 3
14	19	3	4, 5
15	22	3	0-3

In some implementations, for a two codeword transmission for the case when a non-fallback DCI is used and dmrs-Type=2; and the maximum number of front-loaded DMRS symbols is limited to a single symbol, e.g., RRC configured value of maxLength=1, and the configured bit-width of the antenna port indication field in DCI is 0-bits, the

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UE is expected to assume that the default indication is value 0 from the two-codeword column of Table 4. In some implementations, if the configured bit-width of the antenna port indication field in DCI is 1-bits, the UE is expected to assume that bit 0 corresponds to value 0 and bit 1 corresponds to Value 1 from the two-codeword column Table 4.

A gNB can provide an uplink DMRS antenna port indication. In some implementations, for the case when a non-fallback Uplink DCI is used and dmrs-Type=1; and the maximum number of front loaded DMRS symbols is limited to a single symbol, e.g., RRC configured value of maxLength=1, and transform precoding is enabled; and the configured bit-width of the antenna port indication field in DCI is 2-bits, the indication can be in accordance with Table 9. The non-fallback DCI can be based on a DCI format 0₁ or a new DCI format.

TABLE 9

Antenna port(s), transform precoder is enabled, dmrs-Type = 1, maxLength = 1		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0
1	2	1
2	2	2
3	2	3

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 0-bit, the UE is expected to assume the default value of the antenna port indication is value 0 from Table 9. In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 1-bit, the UE is expected to assume that bit 0 corresponds to value 0 of Table 9 and bit 1 corresponds to value 2 of Table 9.

In some implementations, for the case when non-fallback Uplink DCI is used and dmrs-Type=1; and the maximum number of front-loaded DMRS symbols is limited to a single symbol e.g., RRC configured value of maxLength=1, and transform precoding is disabled; and the configured bit-width of the antenna port indication field in DCI is 3-bits, the antenna port indication value can be parsed in accordance with Tables 10, 11, 12, or 13 to determine a DMRS antenna port configuration.

TABLE 10

Antenna port(s), transform precoding is disabled, dmrs-Type = 1, maxLength = 1, rank = 1		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0
1	1	1
2	2	0
3	2	1
4	2	2
5	2	3
6-7	Reserved	Reserved

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TABLE 11

Antenna port(s), transform precoding is disabled, dmrs-Type = 1, maxLength = 1, rank = 2		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0, 1
1	2	0, 1
2	2	2, 3
3	2	0, 2
4-7	Reserved	Reserved

TABLE 12

Antenna port(s), transform precoding is disabled, dmrs-Type = 1, maxLength = 1, rank = 3		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-2
2-7	Reserved	Reserved

TABLE 13

Antenna port(s), transform precoding is disabled, dmrs-Type = 1, maxLength = 1, rank = 4		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-3
2-7	Reserved	Reserved

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 0-bit, the UE is expected to assume that the default indication is value 2 of Table 10, value 1 of Table 11, and value 0 of Tables 12-13 for ranks 1-4 respectively. In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 1-bit, the UE is expected to assume the following: for rank 1: bit 0 corresponds to value 2 and bit 1 corresponds to value 4 in Table 10; for rank 2, bit 0 corresponds to value 1 and bit 1 corresponds to value 2 in Table 10; and for ranks 3 and 4, bit 0 corresponds to value to 0 in Tables 12-13 and bit 1 corresponds to a reserved value.

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 2-bits, the UE is expected to assume the following: for rank 1, values 0-3 corresponds to values 2-5 in Table 10; for rank 2, values 0-3 corresponds to values 0-3 in Table 11; and for ranks 3 and 4, value 0 corresponds to value to 0 in Tables 12-13 and other values correspond to a reserved value.

In some implementations, for the case when a non-fallback Uplink DCI is used and dmrs-Type=2; and the maximum number of front-loaded DMRS symbols is limited to a single symbol e.g., RRC configured value of max-Length=1, and transform precoding is disabled; and the configured bit-width of the antenna port indication field in

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DCI is 4-bits, the antenna port indication value can be parsed in accordance with Tables 14, 15, 16, or 17 to determine a DMRS antenna port configuration.

TABLE 14

Antenna port(s), transform precoding is disabled, dmrs-Type = 2, maxLength = 1, rank = 1		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0
1	1	1
2	2	0
3	2	1
4	2	2
5	2	3
6	3	0
7	3	1
8	3	2
9	3	3
10	3	4
11	3	5
12-15	Reserved	Reserved

TABLE 15

Antenna port(s), transform precoding is disabled, dmrs-Type = 2, maxLength = 1, rank = 2		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0, 1
1	2	0, 1
2	2	2, 3
3	3	0, 1
4	3	2, 3
5	3	4, 5
6	2	0, 2
7-15	Reserved	Reserved

TABLE 16

Antenna port(s), transform precoding is disabled, dmrs-Type = 2, maxLength = 1, rank = 3		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-2
1	3	0-2
2	3	3-5
3-15	Reserved	Reserved

TABLE 17

Antenna port(s), transform precoding is disabled, dmrs-Type = 2, maxLength = 1, rank = 4		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-3
1	3	0-3
2-15	Reserved	Reserved

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 0-bit, the UE is expected to assume that the default indication corresponds to value 6 in Table 14, value 3 in Table 15, and value 1 in Tables 16-17.

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 1-bit, the UE is expected to assume the following: for rank 1, bit 0 corresponds to value to 6 and bit 1 corresponds to value 8 in Table 14; for rank 2, bit 0 corresponds to value to 3 and bit 1 corresponds to value 4 in Table 15; for rank 3, bit 0 corresponds to value to 1 and bit 1 corresponds to value 2 in Table 16; and for rank 4, bit 0 corresponds to value to 0 and bit 1 corresponds to value 1 in Table 17.

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 2-bits, the UE is expected to assume the following: for rank 1, values 0-3 corresponds to values 6-9 in Table 14; for rank 2, values 0-3 corresponds to values 1, 3, 4, 5 in Table 15; and for ranks 3 and 4, values 0-3 corresponds to values 0-3 in Table 16-17.

In some implementations, if the configured bit-width of the antenna port indication field in the DCI is 3-bits, the UE is expected to assume the following: for rank 1: values 0-7 corresponds to values 2, 3, 6-11 in Table 14; and for ranks 2, 3, and 4, values 0-7 correspond to values 0-7 in Tables 15-17.

FIG. 11 illustrates an example of a process for determining an antenna port configuration. At 1105, the UE can receive configuration information that determines a bit-width value for an antenna port indication field of a DCI format. The DCI format can be as described by a current or future wireless communication standard. Receiving the configuration information can include receiving the configuration information via higher layer signaling such as RRC signaling. The bit-width value can be an integer value, e.g., 0, 1, 2, 3, 4, 5, or 6. Other values are possible. If conditions change, the UE can receive subsequent configuration information that indicates a different value for the bit-width value. In some implementations, the lack of a particular configuration parameter in the configuration information can indicate that the bit-width value is zero, e.g., 0-bit. In some implementations, the configuration information includes a bit-width indicator if it is not zero.

At 1110, the UE can receive a DCI message that is in accordance with the DCI format. The UE can monitor the PDCCH for a DCI message. At 1115, the UE can determine an antenna port configuration for one or more uplink or downlink reference signals such as DMRS based on the DCI message, the configuration information, bit-width value, and one or more mapping tables. Determining the antenna port configuration at 1115 can include determining the antenna port configuration based on an antenna port indication in the DCI message and a mapping between antenna port indication values and antenna port numbers. Determining the

antenna port configuration at 1115 can include determining whether there is an antenna port indication in the DCI message based on the bit-width value. Determining the antenna port configuration at 1115 can include using a predetermined antenna port configuration based on a determination that the bit-width value is a zero value, wherein the zero value indicates that the DCI message does not include the antenna port indication field.

At 1120, the UE can transmit or receive the one or more reference signals based on the determined antenna port configuration. In some implementations, the DCI message includes control information to cause the UE to receive one or more demodulation reference signals transmitted by one or more antennas. In some implementations, the UE can use the one or more demodulation reference signals to process a PDSCH transmission. In some implementations, the DCI message includes control information to cause the UE to transmit one or more demodulation reference signals using one or more antennas. In some implementations, the gNB can use the one or more demodulation reference signals to process a PUSCH transmission from the UE.

A technique for DMRS antenna port indication for NR URLLC can include transmitting, receiving, or using a configuration of an antenna port indication field in a DCI message. The technique can be performed by a device such as a gNB or a UE. In some implementations, the configuration of the number of bits for an antenna port indication field in a DCI message is performed via RRC signaling. In some implementations, the configuration is for an uplink URLLC transmission. In some implementations, the configuration is for a downlink URLLC transmission. The configurable antenna port indication field can be part of a new DCI format introduced for URLLC. In some implementations, the configured bit-width of the antenna port indication field in the DCI can 0, 1, 2, 3, 4, 5, or 6 bits for downlink. In some implementations, the configured bit-width of the antenna port indication field in the DCI can 0, 1, 2, 3, 4, 5, or 6 bits for uplink. In some implementations, subsets of values from existing DMRS antenna port indication tables can be mapped to the configured bit-width of the new antenna port indication field for NR URLLC based on technique described above for Tables 1-17.

Another technique for provide antenna port information includes encoding, for transmission to a UE, an indication of a number of bits in a DMRS antenna port indication field of a DCI; and encoding, for transmission to the UE, the DCI including the DMRS antenna port indication field based on the indication. The technique can be performed by a base station such as a gNB. In some implementations, the indication is transmitted via RRC signaling. In some implementations, the DCI is in accordance with DCI format 0_1 or 1_1. The technique can include transmitting a downlink signal based on the DMRS antenna port indication field. The technique can include receiving an uplink signal based on the DMRS antenna port indication field. In some implementations, the DMRS antenna port indication field is for URLLC communication. In some implementations, DMRS antenna port indication field indicates an antenna port for an uplink or downlink signal.

A wireless communication technique can include receiving an indication of a number of bits in a DMRS antenna port indication field of a DCI; receiving the DCI including the DMRS antenna port indication field; and determining an antenna port configuration for an uplink or downlink signal based on the DMRS antenna port indication and the indication. In some implementations, the indication is transmitted via RRC signaling. In some implementations, the DCI is

in accordance with a DCI format 0_1 or 1_1. Implementations can include receiving a downlink signal based on the determined antenna port configuration. Implementations can include transmitting or causing to transmit an uplink signal based on the determined antenna port configuration. In some implementations, the DMRS antenna port indication field is for URLLC communication.

These and other techniques can be performed by an apparatus that is implemented in or employed by one or more types of network components, user devices, or both. In some implementations, one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more of the described techniques. An apparatus can include one or more processors and one or more computer-readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform one or more of the described techniques.

The methods described here may be implemented in software, hardware, or a combination thereof, in different implementations. In addition, the order of the blocks of the methods may be changed, and various elements may be added, reordered, combined, omitted, modified, and the like. Various modifications and changes may be made as would be obvious to a person skilled in the art having the benefit of this disclosure. The various implementations described here are meant to be illustrative and not limiting. Many variations, modifications, additions, and improvements are possible. Accordingly, plural instances may be provided for components described here as a single instance. Boundaries between various components, operations and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of claims that follow. Finally, structures and functionality presented as discrete components in the example configurations may be implemented as a combined structure or component.

The methods described herein can be implemented in circuitry such as one or more of: integrated circuit, logic circuit, a processor (shared, dedicated, or group) and/or memory (shared, dedicated, or group), an Application Specific Integrated Circuit (ASIC), a field-programmable device (FPD) (e.g., a field-programmable gate array (FPGA), a programmable logic device (PLD), a complex PLD (CPLD), a high-capacity PLD (HCPLD), a structured ASIC, or a programmable SoC), digital signal processors (DSPs), or some combination thereof. Examples of processors can include Apple A-series processors, Intel® Architecture Core™ processors, ARM processors, AMD processors, and Qualcomm processors. Other types of processors are possible. In some implementations, the circuitry may execute one or more software or firmware programs to provide at least some of the described functionality. The term “circuitry” may also refer to a combination of one or more hardware elements (or a combination of circuits used in an electrical or electronic system) with the program code used to carry out the functionality of that program code. In these embodiments, the combination of hardware elements and program code may be referred to as a particular type of circuitry. Circuitry can also include radio circuitry such as a transmitter, receiver, or a transceiver.

A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made. Elements of one or more implementations

may be combined, deleted, modified, or supplemented to form further implementations. As yet another example, the logic flows depicted in the figures do not require the particular order shown, or sequential order, to achieve desirable results. In addition, other steps may be provided, or steps may be eliminated, from the described flows, and other components may be added to, or removed from, the described systems. Accordingly, other implementations are within the scope of the following claims.

What is claimed is:

1. A method comprising:

receiving configuration information associated with an antenna port indication field of a downlink control information (DCI) format;
determining that a bit-width value for the antenna port indication field of the DCI format is equal to zero based at least on the configuration information;
receiving a DCI message that is in accordance with the DCI format; and
determining, based at least on the bit-width value being equal to zero, to use a default value for an antenna port configuration, instead of determining a value for the antenna port configuration based on the antenna port indication field in the DCI message; and
receiving one or more demodulation reference signals using the antenna port configuration based at least in part on a result of the determination to use the default value for the antenna port configuration.

2. The method of claim 1, wherein determining the antenna port configuration comprises determining an antenna port number for the antenna port configuration based on the default value and a mapping between antenna port indication values and antenna port numbers.

3. The method of claim 2, wherein the default value corresponds to an entry of the mapping between the antenna port indication values and the antenna port numbers.

4. The method of claim 1, wherein determining the antenna port configuration comprises using a predetermined antenna port configuration based on a determination that the bit-width value of the antenna port indication field is equal to zero.

5. The method of claim 1, wherein the configuration information indicates that the antenna port indication field is not configured for the DCI format.

6. The method of claim 1, wherein receiving the configuration information comprises receiving radio resource control (RRC) signaling that indicates the configuration information.

7. The method of claim 1, wherein receiving the one or more demodulation reference signals comprises receiving the one or more demodulation reference signals using one or more demodulation reference signal ports associated with the default value.

8. An apparatus comprising:

one or more processors; and
memory storing instructions that, when executed, are configured to cause the one or more processors to perform operations comprising:
receiving configuration information associated with an antenna port indication field of a downlink control information (DCI) format;
determining that a bit-width value for the antenna port indication field of the DCI format is equal to zero based at least on the configuration information;
receiving a DCI message that is in accordance with the DCI format;

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determining, based at least on the bit-width value being equal to zero, to use a default value for an antenna port configuration, instead of determining a value for the antenna port configuration based on the antenna port indication field in the DCI message; and
 receiving one or more demodulation reference signals using the antenna port configuration based at least in part on a result of the determination to use the default value for the antenna port configuration.

9. The apparatus of claim 8, wherein determining the antenna port configuration comprises determining an antenna port number for the antenna port configuration based on the default value and a mapping between antenna port indication values and antenna port numbers.

10. The apparatus of claim 9, wherein the default value corresponds to an entry of the mapping between the antenna port indication values and the antenna port numbers.

11. The apparatus of claim 8, wherein determining the antenna port configuration comprises using a predetermined antenna port configuration based on a determination that the bit-width value of the antenna port indication field is equal to zero.

12. The apparatus of claim 8, wherein the configuration information indicates that the antenna port indication field is not configured for the DCI format.

13. The apparatus of claim 8, wherein receiving the configuration information comprises receiving radio resource control (RRC) signaling that indicates the configuration information.

14. The apparatus of claim 8, wherein receiving the one or more demodulation reference signals comprises receiving the one or more demodulation reference signals using one or more demodulation reference signal ports associated with the default value.

15. An apparatus comprising:

one or more processors; and

memory storing instructions that, when executed, are configured to cause the one or more processors to perform operations comprising:

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transmitting configuration information associated with an antenna port indication field of a downlink control information (DCI) format;

determining that a bit-width value for the antenna port indication field of the DCI format is equal to zero based at least on the configuration information;

transmitting a DCI message that is in accordance with the DCI format;

determining a default value for an antenna port configuration based at least on the bit-width value being equal to zero, instead of determining a value for the antenna port configuration based on the antenna port indication field in the DCI message; and

transmitting one or more demodulation reference signals using the antenna port configuration based at least in part on a result of the determination to use the default value for the antenna port configuration.

16. The apparatus of claim 15, wherein determining the antenna port configuration comprises determining an antenna port number for the antenna port configuration based on the default value and a mapping between antenna port indication values and antenna port numbers.

17. The apparatus of claim 16, wherein the default value corresponds to an entry of the mapping between the antenna port indication values and the antenna port numbers.

18. The apparatus of claim 15, wherein determining the antenna port configuration comprises using a predetermined antenna port configuration based on a determination that the bit-width value of the antenna port indication field is equal to zero.

19. The apparatus of claim 15, wherein the configuration information indicates that the antenna port indication field is not configured for the DCI format.

20. The apparatus of claim 15, wherein receiving the configuration information comprises receiving radio resource control (RRC) signaling that indicates the configuration information.

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